

Priced Out: Financing Frictions and Startup Mortality^{*}

Rodrigo Pacheco[†]

Columbia University

Madhav Malhotra[‡]

Columbia University

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Abstract

Startups raise capital in a sequence of rounds, and each round exposes them to financing risk, the possibility that a viable firm fails because it cannot raise its next round on acceptable terms. This paper asks how much startup failure financing risk causes, and what determines it. We develop a dynamic model of staged equity financing in which a firm with stochastic fundamentals must repeatedly return to the market to raise more capital, and we summarize financing frictions as a wedge between the price of outside capital and its frictionless level. Two results organize the analysis. First, because equity is repriced at every round, repricing substitutes for coordination among investors, so the equilibrium is unique and staged equity does not admit self-fulfilling failure. Second, a firm fails when its fundamentals fall below the level at which investors are willing to fund another round, and the wedge raises that level: firms that would survive in a frictionless market fail instead. At the estimated wedge, 11.6% of startup failures are attributable to financing frictions rather than to fundamentals, and six in ten of these firms would survive to exit if the wedge were removed.

Keywords: Startups, venture capital, financing frictions

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[†]Email: rcp2157@columbia.edu.

[‡]Email: mm6112@columbia.edu.

[§]Results are subject to change. Please do not cite or distribute.

1 Introduction

Startups raise capital in stages. Rather than financing the entire development process at once, investors provide enough capital to reach the next stage and decide whether to continue once more information has arrived. As a result, a startup can fail not only because its fundamentals deteriorate, but also because it cannot raise its next round on terms that make continuation worthwhile. This paper asks how much startup failure is caused by this financing risk.

Why can a firm that is worth continuing fail to raise its next round? One possibility is coordination: investors may refuse to fund the firm because they expect future investors to do the same. Another is that raising outside capital is itself costly, so the terms offered by a new investor can make continuation unattractive even when funding the firm creates positive surplus. We distinguish these two mechanisms and quantify the second.

We do so with a model of staged equity financing and a panel of venture-backed firms from PitchBook. In the model, firms repeatedly raise capital either from their current owners or from new investors. When a new investor is required, it captures a fraction of the surplus created by the round. We summarize this financing friction with a single wedge between the price of outside capital and the price at which a new investor would just break even. Our baseline estimate of the wedge is 3%. At this estimate, the wedge accounts for 11.6% of startup failures. If the wedge were eliminated, about 60% of the firms whose failure is attributable to it would eventually survive to a successful exit.

The model consists of a firm that pays a single dividend at a random date and must raise capital every time a financing need arrives before then. The dividend is the firm's fundamentals. Its size is observable by every investor and evolves as a geometric Brownian motion, with a drift and a volatility. Financing needs arrive at a constant rate, and each one forces the firm to raise a round or go bankrupt. Investors are risk neutral, discount the future at a common rate, and are in unlimited supply.

A round is funded either by the owners or by a new investor who bargains for a stake, and the wedge determines how much of the surplus the new investor obtains. Rounds are of two types. At an inside round, money is all the firm needs, and the current owners decide whether to put it in or let the firm fail. At a market round, the firm needs an input its current owners cannot provide, and it must sell equity to a new investor who has it, and that investor decides whether to fund the round or let the firm fail. Failure means bankruptcy, with every equity claim worth zero.

Because the terms of every round are set after the dividend is observed, the equilibrium is unique and no firm fails only because investors expect it to fail. This rules out the first channel. The contrast with debt makes the mechanism clear. When debt is rolled over at fixed terms, as in [He and Xiong \(2012a\)](#), a maturing creditor's payoff depends on whether the

creditors maturing after it roll over as well, and the resulting rat race makes creditors run at fundamentals at which they would otherwise have stayed. Here the stake a new investor receives adjusts to what the firm is worth once the round is funded, and that value already reflects the rounds still to come. A new investor is therefore compensated for the risk of future rounds through its stake rather than exposed to it at a fixed price.

We show that the equilibrium takes the form of a threshold. At every financing need, the round is funded if and only if the dividend is above a threshold, and the wedge raises that threshold.

The mechanism is dynamic. The wedge does not only affect how the surplus of the current round is divided. The owners anticipate that new investors will capture part of the surplus at every future round as well, so a larger wedge reduces the value of equity today. This lowers the value of funding the firm whenever a financing need arrives and raises the dividend required for financing to continue. A small friction at each round can therefore have a substantially larger effect on the firm's continuation value because it is expected to be paid repeatedly over the firm's remaining life.

The market without the wedge maximizes total surplus, the sum of the payoffs of the founder and of every investor. A stake is a transfer between players with the same preferences, so total surplus does not depend on how the equity is split, and the funding policy that maximizes it funds the firm if and only if its dividend is above the threshold of the market without the wedge. We refer to this policy as the efficient benchmark and to its threshold as the efficient threshold. A firm whose dividend lies between the two thresholds when a financing need arrives is worth funding and only fails because of the wedge. Therefore, we refer to these failures as inefficient.

We then ask what fraction of all failures are inefficient. In the stationary state of a population of such firms, the share of failures that occur between the two thresholds has a closed form that depends only on the ratio of the two thresholds, on the drift and the volatility of the dividend, and on how often financing needs and the exit arrive, regardless of the distribution of firms at entry. Removing the wedge does not save all of these firms, since some later fall below the efficient threshold and fail anyway. The share of failures that removing the wedge prevents has a closed form of the same shape, and the ratio of the two shares is the fraction of the saved firms that survive to exit.

The data are a panel of venture-backed firms from PitchBook. The panel covers 41,722 firms headquartered in the United States that raised their first priced round between 2005 and 2018, and it follows each of them through October 2025, by which time the firm has exited through an initial public offering, an acquisition or a buyout, has failed, or is still active. We classify a round as a market round when a new investor leads it and as an inside round when every lead is a returning investor. The lead investor writes the largest check and sets the terms

of the deal, the valuation and the stake, which the other participants accept, so a round led by a returning investor is priced by the owners and a round led by a newcomer is priced by an outsider who bargains with them, as in the model.

The difference in financing terms between market and inside rounds identifies the wedge. In the model, conditional on the state of the firm, the owners retain a smaller share after a market round because the new investor captures part of the surplus. Empirically, we compare retained ownership across market and inside rounds while absorbing differences in firm stage, market conditions, and persistent firm characteristics with round-number, year-by-sector, and firm fixed effects. The estimate implies that a new investor captures approximately 3% of the surplus created by a market round.

The remaining parameters govern the evolution of the dividend. They cannot be read directly from observed exit values because exit values are selected: only firms that survive all previous financing needs reach exit. We therefore estimate the drift and the volatility of the dividend by simulated method of moments. The model reproduces this endogenous selection, allowing us to choose the parameters so that the growth and dispersion of exit values in simulated data match those observed in PitchBook.

Related Literature. This paper contributes to four strands of the literature.

First, this paper relates to the literature on financing risk in staged equity financing (see [Nanda and Rhodes-Kropf, 2017](#); [Parry, 2025](#); [Chen, 2026](#)). [Nanda and Rhodes-Kropf \(2017\)](#) define financing risk as the possibility that a viable firm fails because investors expect future investors not to fund it, and obtain it in a model in which the firm's exit value depends on whether it can raise funds. They also show, in an appendix, that the risk disappears in a simpler model when the stake sold at each round adjusts to the firm's prospects, because a firm worth funding can always offer a larger stake. We prove the general version of that observation. With stochastic fundamentals that are exogenous to financing, a new investor at every market round, and stakes set by bargaining rather than by competition, the equilibrium is unique and coordination among investors cannot cause failure. What survives repricing is the price of outside capital, which their competitive investors do not charge.

[Parry \(2025\)](#) is the closest quantitative work. He estimates a model in which entrepreneurs and investors meet in a frictional matching market and investors stage capital to limit losses from entrepreneurs who hide failure. His financing risk is the risk of not finding an investor, and ours is the price that the investor found charges. His estimate is a share of startups and ours a share of failures, so the two are not directly comparable, but they are complements: a thin market for outside capital lowers his matching rate and raises our wedge.

[Chen \(2026\)](#) constructs a text-based, firm-level measure of financing risk, the expectation that a startup will not be able to raise its next round, and shows that it reduces innovation and

growth and raises failure even among recently funded firms. In our model that probability is an equilibrium object, the chance that the next financing need arrives while the dividend is below the funding threshold.

Second, this paper relates to the literature on staged financing and hold-up. Staged financing itself is the response to two problems, hold-up by a founder who cannot commit her human capital to the venture (see [Hart and Moore, 1994](#); [Neher, 1999](#)) and learning about a project whose success and completion time are uncertain ([Bergemann and Hege, 2005](#)). We take staging as given and study the market it creates, in which the hold-up runs the other way, of the firm by the investor who supplies the next round rather than of the investor by the founder. [Hellmann and Thiele \(2015\)](#) model this hold-up in a market with two stages, in which the later investor no longer needs the earlier one, and derive its effect on valuations and exit rates. Our market rounds are a stationary, repeated version of it, and we measure its cost. [Admati and Pfleiderer \(1994\)](#) study how an inside investor prices the rounds that follow its entry and show that a fixed-fraction contract removes its incentive to misprice them, which is the reference point for our treatment of inside rounds as priced at the equity value.

Third, the wedge itself has microfoundations in the literature on the market for venture capital. [Michelacci and Suarez \(2004\)](#), [Inderst and Müller \(2004\)](#), [Silveira and Wright \(2016\)](#) and [Ewens, Gorbenko and Korteweg \(2022\)](#) model the market as one of search and matching between entrepreneurs and investors, in which the surplus of a match is split by bargaining, so the share an investor obtains reflects how long the founder would take to find another one. In [Inderst and Müller \(2004\)](#) the bargaining power of investors is itself determined by the relative scarcity of venture capital. [Jovanovic and Szentes \(2013\)](#) obtain rents for investors without search frictions, from the scarcity of venture capitalists in a competitive equilibrium. We do not model where the wedge comes from. We take it as the share of the surplus that a new investor captures at every market round, identify it from the difference between the stakes sold at market and at inside rounds, and measure what it costs in failures. [Ewens, Gorbenko and Korteweg \(2022\)](#) estimate the bargaining power of investors from first rounds, whereas our wedge is measured on later rounds, relative to the rounds that the owners price themselves.

Finally, this paper relates to the literature on rollover risk and dynamic runs (see [Acharya, Gale and Yorulmazer, 2011](#); [He and Xiong, 2012a,b](#); [Brunnermeier and Oehmke, 2013](#); [Schroth, Suarez and Taylor, 2014](#); [He and Milbradt, 2016](#)). Our model is built on [He and Xiong \(2012a\)](#), in which a firm with a time-varying fundamental repeatedly rolls over debt at terms that do not adjust to the fundamental, so each maturing creditor's decision depends on the decisions of the creditors who come after, and runs occur at fundamentals at which creditors would otherwise stay. In [He and Milbradt \(2016\)](#) and [Brunnermeier and Oehmke \(2013\)](#) the fragility likewise comes from outstanding claims with fixed terms that future issuance can dilute. That

equity claims are not exposed to runs is an old observation (Jacklin, 1987), and we show what it implies for staged financing.

Outline. The rest of the paper is organized as follows. Section 2 develops the model of staged equity financing, defines the equilibrium and shows that it is unique and takes the form of a funding threshold. Section 3 defines the efficient benchmark, shows that the market attains it when the wedge is removed, and derives in closed form the share of failures the wedge causes and the share it would be possible to prevent. Section 4 describes the data and maps them to the model. Section 5 estimates the parameters, and Section 6 reports the cost of the wedge at the estimates. Section 7 concludes. Proofs are in Appendix A.

2 Model

This section develops a model of staged equity financing. A single firm with a stochastic dividend must repeatedly raise capital to stay alive, and each round is funded either by the firm's current owners or by a new investor from the market. Financing frictions enter through a single parameter, a wedge θ between the price of outside capital and its competitive level, the price at which a new investor just breaks even. At every round funded by a new investor, that investor captures a share θ of the surplus the round creates. The next subsection describes the firm and its investors. The financing game that follows specifies who moves at each round, what they can do, and on what terms. We then define the equity value under an arbitrary strategy profile, restrict attention to symmetric Markov perfect equilibria, and show that the equilibrium is unique and takes the form of a funding threshold. The firm raises another round if and only if its dividend exceeds a cutoff, and the wedge raises that cutoff.

2.1. Environment

The economy is set in continuous time over an infinite horizon. We study a single firm in partial equilibrium, with investors who are risk neutral, discount the future at rate ρ , and are in unlimited supply.

The firm generates no cash flow until a random date, at which it pays a single, one-time dividend to its equity holders and is dissolved. The size of the dividend y_t is publicly observable and follows a geometric Brownian motion

$$\frac{dy_t}{y_t} = \mu dt + \sigma dZ_t$$

where Z_t is a standard Brownian motion, μ is the drift and σ the volatility. A higher y_t means

a larger payout to whoever holds equity when it is paid. Modeling the dividend as a process, rather than as a fixed number, captures the nature of a startup. What the firm will be worth when its value is realized depends on the outcome of a development process that is still under way, so the firm an investor sees today can look very different one year from now.

Two independent Poisson processes govern the events in the firm's life. The first is a payout shock, which arrives at rate ϕ . When it arrives, the firm pays the dividend y_t to its equity holders, pro rata, and is dissolved. The second is a financing need, which arrives at rate λ and forces the firm to raise

$$x(y_t) = \alpha y_t + x_0$$

in new equity or go bankrupt, in which case every equity claim pays zero. Both processes are independent of each other and of Z_t . The payout shock is the date at which the firm's value is realized and paid out. The financing need captures the fact that a startup burns cash between rounds and must return to the market at dates it does not control. The development process is uncertain, and so is the amount of money it takes to keep the firm alive until the payout. A setback, a longer trial, or a product that needs one more iteration before it sells all mean that the cash raised at the last round runs out earlier than planned, and the firm must raise again or fail. Modeling financing needs as arrivals of a Poisson process captures this uncertainty in the simplest way, with a single parameter for how often the firm has to return to the market. The size of the round has a fixed component $x_0 > 0$ and a component αy_t , with $\alpha \geq 0$, that scales with the firm, so we allow for the possibility that larger firms raise larger rounds.

2.2. The Financing Game

Each financing need is a round, and rounds can be one of two types. With probability $1 - p$ the round is an inside round, in which the investors who currently hold the firm's equity decide whether to put in $x(y)$ themselves or let the firm fail. With probability p the round is a market round, in which the current holders cannot fund it and the firm must raise $x(y)$ from a new outside investor. The type of a round is drawn independently of everything else, and p measures how often the firm has to go back to the market.

The two types of round capture the idea that a round is not only a transfer of money. As the development process unfolds, the firm may find that it needs something its current owners cannot provide, expertise in a market it is entering, a network of customers, or operating experience it lacks, and the way to bring it in is to sell equity to an investor who has it. An inside round is one in which money is all the firm needs, and its owners can supply it. A market round is one in which the firm must go to the market, and it raises the capital and the missing input in the same transaction. Which of the two a given round turns out to be is a feature of the development process, not a choice, which is why we model it as an independent

draw.

The players are the founder and a sequence of outside investors, indexed by $i = 0, 1, 2, \dots$, with investor 0 the founder, who owns the firm at time 0. Let T_i be the arrival time of the i -th financing need. If round i is a market round, investor i is matched to the firm and called to move. If round i is an inside round, investor i never enters. An investor is called to move as a new investor at most once. After funding a market round it joins the current owners, the investors holding the firm's equity, who act as a coalition at every inside round. We treat this coalition as a single player, the owners.

At round T_i the player called to fund, whether the owners or investor i , observes the public history h_{T_i} , which records the path of the dividend and the dates, types and outcomes of all previous rounds, and chooses one of two actions, fund or walk. Walk means the same at every round. The round fails, the firm goes bankrupt, and every equity claim pays zero. Fund puts $x(y_{T_i})$ into the firm, and its terms depend on the type of the round.

At an inside round the owners put in $x(y_{T_i})$ and keep all of the firm's equity. At a market round investor i pays $x(y_{T_i})$ for a stake s_i of the firm's equity, and the stakes of the owners are diluted by the factor $1 - s_i$. The stake s_i is set by Nash bargaining between investor i and the owners, with the wedge θ as the investor's bargaining weight. Because the bargaining outcome depends on what the firm is worth once the round is funded, and that value depends in turn on how future rounds will be played, we solve the bargaining problem in the next subsection, after defining the equity value for a given strategy profile.

The game ends at whichever of two events comes first. At the payout time τ_ϕ , the equity holders split y_{τ_ϕ} pro rata. At the first unfunded round, the firm goes bankrupt and every equity claim pays zero.

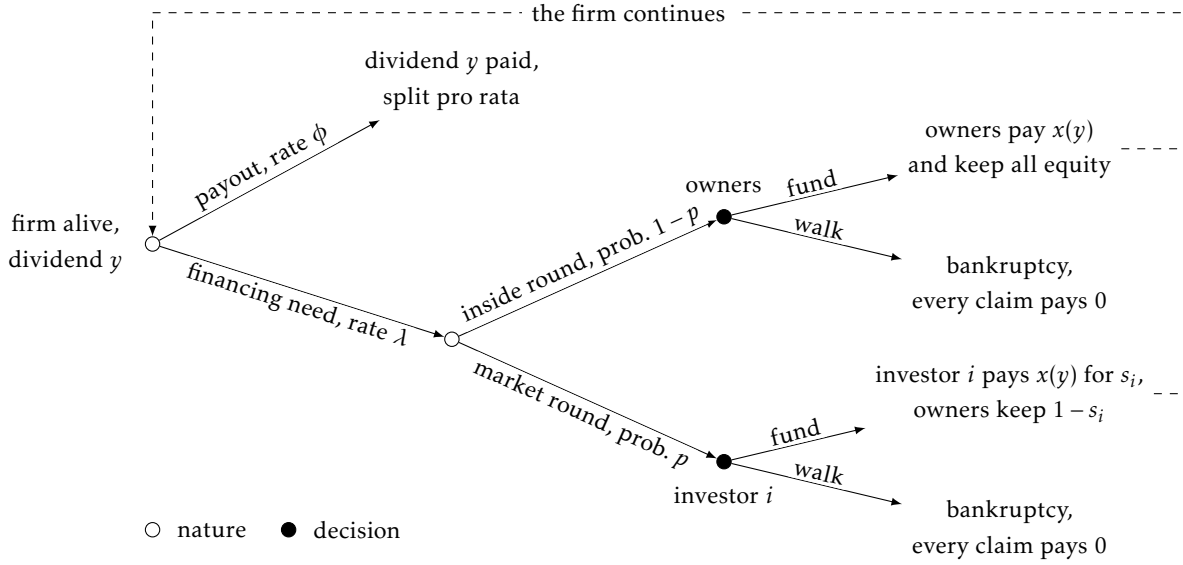
A pure strategy maps public histories into actions. Investor i moves only at its market round and the owners at every inside round, so

$$\pi_i : h_{T_i} \mapsto \{\text{fund}, \text{walk}\}, \quad \pi_o : h_{T_i} \mapsto \{\text{fund}, \text{walk}\}$$

where π_i is the strategy for investor i and π_o is the strategy for the owners. We will refer to a strategy profile as $\pi = (\pi_o, (\pi_i)_{i \geq 1})$.

Figure 1 summarizes the game.

Figure 1: The financing game



2.3. Valuation and the Terms of a Market Round

Fix a strategy profile π . Under π the play of the game from any history is determined, so the firm's equity has a well defined value, which we will refer to as the equity value. It is what the owners weigh against the cost of an inside round, and what investor i and the owners bargain over at a market round, so we define it first and then return to the bargaining problem.

Consider the investors holding the firm's equity at time t , after history h_t . Between t and the payout they are diluted at every market round that is funded, they pay the cost of every inside round that is funded, and they receive nothing if any round is left unfunded. Therefore, the equity value at t is

$$W^\pi(h_t) = \mathbb{E} \left[e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} D(t, \tau_\phi) S(t, \tau_\phi) - \sum_{\substack{\text{inside } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} x(y_{T_j}) D(t, T_j) S(t, T_j) \mid h_t \right] \quad (1)$$

where D is the dilution factor and S the survival factor

$$D(t, u) = \prod_{\substack{\text{market } j \\ t < T_j \leq u}} (1 - s_j), \quad S(t, u) = \prod_{\substack{\text{inside } j \\ t < T_j \leq u}} \mathbb{1}_{\{\pi_o(h_{T_j}) = \text{fund}\}} \cdot \prod_{\substack{\text{market } j \\ t < T_j \leq u}} \mathbb{1}_{\{\pi_i(h_{T_j}) = \text{fund}\}}$$

The first term in Eq. (1) is the discounted dividend. The dilution factor accounts for the fact that the holders at t give up a share s_j of the firm at every market round funded between t and u , and pay nothing for it. The survival factor equals one if every round between t and u is funded under π and zero otherwise, which is the sense in which every round exposes the firm

to financing risk. The second term is the cost of the inside rounds between t and the payout. The owners bear it in full at the time, and the equity holders at t bear the fraction $D(t, T_j)$ of it that matches the equity they still hold when that round happens. The expectation is over the stochastic dividend, the Poisson processes governing the payout and financing needs and the round types, conditional on the history h_t .

We now return to the terms of a market round. Let $h_{T_i}^f$ denote the history in which the round at T_i has just been funded, so that $W^\pi(h_{T_i}^f)$ is the equity value once investor i 's capital is in. If the round is funded, the owners keep a share $1 - s_i$ of that value and investor i receives a share s_i in exchange for $x(y_{T_i})$. If the two sides do not agree, the round is unfunded, the firm goes bankrupt, and both sides receive zero. The stake s_i solves the Nash bargaining problem¹

$$\max_{s \in [0,1]} \underbrace{\left[(1-s) W^\pi(h_{T_i}^f) \right]^{1-\theta}}_{\text{owners}} \underbrace{\left[s W^\pi(h_{T_i}^f) - x(y_{T_i}) \right]^\theta}_{\text{investor } i}$$

If the round is funded, it creates a total surplus of $W^\pi(h_{T_i}^f) - x(y_{T_i})$, and the problem has a solution with both sides willing to fund if and only if this surplus is nonnegative. The Nash solution then gives investor i a fraction θ of the total surplus, and the owners receive the rest:

$$(1 - s_i) W^\pi(h_{T_i}^f) = (1 - \theta) \left(W^\pi(h_{T_i}^f) - x(y_{T_i}) \right), \quad s_i W^\pi(h_{T_i}^f) = x(y_{T_i}) + \theta \left(W^\pi(h_{T_i}^f) - x(y_{T_i}) \right) \quad (2)$$

Investor i receives a claim worth what it paid plus a share θ of the surplus. We can solve the second equation in Eq. (2) for the stake itself:

$$s_i = \theta + (1 - \theta) \frac{x(y_{T_i})}{W^\pi(h_{T_i}^f)} \quad (3)$$

which is a weighted average of one and of the share of the firm that the new capital represents. The two limiting cases show what the wedge does. When $\theta = 1$, the new investor takes the whole firm in exchange for the capital, regardless of the firm's value or of the size of the round, and the owners are left indifferent between funding and bankruptcy. When $\theta = 0$, the new investor receives exactly the share $x(y_{T_i})/W^\pi(h_{T_i}^f)$ that its capital represents, so it pays what its claim is worth. This is the competitive price of outside capital, the price at which competition among new investors for the round would leave the new investor with no

¹The Nash bargaining solution is the limit of a noncooperative game in which the two parties alternate offers indefinitely under the risk that negotiations break down (Binmore, Rubinstein and Wolinsky, 1986). In that limit the disagreement point is the payoff at breakdown, here bankruptcy, and the asymmetric weight θ reflects an asymmetry in the bargaining procedure, such as the time each side waits after making an offer or the probability of being the one to make it.

surplus. Every $\theta > 0$ moves the stake toward one and raises the price of outside capital above its competitive level, which is what makes θ a wedge.

At a market round, the expected payoff to investor i if it funds is

$$\mathbb{E}[U_{i,T_i} \mid h_{T_i}, \pi_i(h_{T_i}) = \text{fund}] = -x(y_{T_i}) + s_i W^\pi(h_{T_i}^f) = \theta \left(W^\pi(h_{T_i}^f) - x(y_{T_i}) \right)$$

Since walking pays zero, investor i 's best response is to fund if and only if the round creates a nonnegative surplus

$$\pi_i(h_{T_i}) = \text{fund} \iff W^\pi(h_{T_i}^f) \geq x(y_{T_i}) \quad (4)$$

where we adopt the convention that a player funds when indifferent.²

At an inside round T_j , the expected payoff to the owners if they fund is

$$\mathbb{E}[U_{o,T_j} \mid h_{T_j}, \pi_o(h_{T_j}) = \text{fund}] = -x(y_{T_j}) + W^\pi(h_{T_j}^f)$$

since they hold all of the firm's equity before and after the round. Walking again pays zero, so the owners' best response is

$$\pi_o(h_{T_j}) = \text{fund} \iff W^\pi(h_{T_j}^f) \geq x(y_{T_j}) \quad (5)$$

Both players apply the same rule. Each weighs the same surplus, $W^\pi(h^f) - x$, against zero. Investor i receives a fraction θ of it and the owners receive all of it, so the wedge changes how the surplus is shared but not its sign. Whether a round is funded therefore does not depend on who is asked to fund it. The next subsection describes the equilibrium concept and shows that the equilibrium is unique and takes the form of a funding threshold.

2.4. Equilibrium

We will restrict attention to Markov strategies, that is, strategies that depend on the history only through the dividend at the round. A Markov strategy for investor i is a measurable funding set $F_i \subseteq (0, \infty)$ such that

$$\pi_i(h_{T_i}) = \text{fund} \iff y_{T_i} \in F_i$$

and a Markov strategy for the owners is a measurable funding set $F^o \subseteq (0, \infty)$ such that

$$\pi_o(h_{T_j}) = \text{fund} \iff y_{T_j} \in F^o$$

²The convention matters only when the surplus is exactly zero and, when $\theta = 0$, at every history, since a new investor with no bargaining weight breaks even whenever it funds.

Investor i funds its market round if and only if the dividend at that round lies in F_i , and the owners fund an inside round if and only if the dividend lies in F^o . Every new investor faces the same problem, so we look for symmetric profiles, in which every investor uses the same funding set F . We will refer to a pair (F, F^o) as a symmetric Markov profile.

We write $W^{F, F^o}(y)$ for the equity value given the funding sets F and F^o . Applying Itô's lemma to Eq. (1), the equity value satisfies the Hamilton-Jacobi-Bellman (HJB) equation

$$\begin{aligned} \rho W^{F, F^o}(y) = & \mu y \frac{dW^{F, F^o}}{dy} + \frac{\sigma^2}{2} y^2 \frac{d^2 W^{F, F^o}}{dy^2} + \underbrace{\phi(y - W^{F, F^o}(y))}_{\text{payout}} \\ & - \lambda(1-p) \underbrace{\left[x(y) \mathbb{1}_{\{y \in F^o\}} + W^{F, F^o}(y) \mathbb{1}_{\{y \notin F^o\}} \right]}_{\text{inside round}} \\ & + \lambda p \underbrace{\left[((1-\theta)(W^{F, F^o}(y) - x(y)) - W^{F, F^o}(y)) \mathbb{1}_{\{y \in F\}} - W^{F, F^o}(y) \mathbb{1}_{\{y \notin F\}} \right]}_{\text{market round}} \end{aligned} \quad (6)$$

The left side is the required return on the equity. On the right, the first two terms are the expected capital gain from the drift and the volatility of the dividend. The third term is the payout, at rate ϕ , at which the equity holders receive y and give up the equity value. The fourth term is an inside round, at rate $\lambda(1-p)$, at which the owners pay $x(y)$ if the dividend lies in F^o and lose everything otherwise. The fifth term is a market round, at rate λp . If the dividend lies in F , the equity holders keep the share $1 - s_i$ of the equity value, which equals $(1 - \theta)(W^{F, F^o}(y) - x(y))$ by Eq. (2), and if it does not, they lose everything.

We can now define the equilibrium.

Definition 1 (Symmetric Markov perfect equilibrium). A symmetric Markov profile (F, F^o) is a symmetric Markov perfect equilibrium if

$$F = \{y : W^{F, F^o}(y) \geq x(y)\}, \quad F^o = \{y : W^{F, F^o}(y) \geq x(y)\}$$

The definition states that F and F^o are a fixed point. A profile (F, F^o) determines the equity value W^{F, F^o} through Eq. (6). Given that equity value, Eqs. (4) and (5) say that a new investor and the owners fund a round exactly when $W^{F, F^o}(y) \geq x(y)$, which defines a funding set for each of them. The profile is an equilibrium when the funding sets it induces in this way are F and F^o themselves. It follows from the definition that the two funding sets are equal in any equilibrium, $F = F^o$. The owners and a new investor fund the firm at exactly the same dividends.

We will refer to the common set F as the funding set and write W for the equilibrium equity value $W^{F, F}$. Imposing $F = F^o = \{y : W(y) \geq x(y)\}$ in Eq. (6), the inside and market round

terms combine and the HJB collapses to

$$\rho W = \mu y W' + \frac{\sigma^2}{2} y^2 W'' + \phi(y - W) + \lambda(1 - p\theta)(W - x(y))^+ - \lambda W \quad (7)$$

where $(u)^+ = \max\{u, 0\}$ and the argument y of W is omitted. The last two terms describe a financing need, which arrives at rate λ . The equity holders lose the equity value, captured by the last term, and if the round creates a positive surplus they recover a fraction of it. They recover all of the surplus at an inside round, which occurs with probability $1 - p$, and a fraction $1 - \theta$ of it at a market round, which occurs with probability p , so the expected fraction recovered is $1 - p\theta$. If the surplus is negative the round is not funded and nothing is recovered.

Two features of Eq. (7) are worth noting. First, the equilibrium is characterized by W alone. Once W is known, the funding set is $F = \{y : W(y) \geq x(y)\}$, and the stake at every market round follows from Eq. (3). Second, the frequency of market rounds and the wedge enter only through the product $p\theta$, the expected share of the surplus that new investors receive at a financing need. A firm that goes to the market almost every round and cedes a small share of the surplus each time is in the same position as a firm that rarely goes to the market and cedes a large share when it does.

To guarantee that the equilibrium is unique whenever it exists, we impose one condition on the parameters.

Assumption 1. $\mu < \rho + \phi + \lambda p\theta$.

Assumption 1 requires that the dividend grow more slowly than the rate at which the equity holders discount it. It ensures that the operator whose fixed point is the equilibrium equity value is a contraction.

Proposition 1 (Uniqueness). *Under Assumption 1, there is at most one symmetric Markov perfect equilibrium.*

Proof: See Appendix A.1.

The contrast with debt runs is instructive. In He and Xiong (2012a), debt is rolled over at fixed terms, so a maturing creditor's payoff from rolling over depends on whether the creditors maturing after it roll over as well. Each creditor's rollover threshold rises with the thresholds of the others, and this rat race leads creditors to run at fundamentals at which they would have stayed absent the coordination problem. Here the terms of every round are set after the dividend is observed and adjust to the equity value, so a new investor is compensated for the risk of future rounds through the stake it receives rather than exposed to it at a fixed price. Repricing substitutes for coordination. The funding set is pinned down by the equity value alone, and no firm fails only because investors expect it to fail.

Proposition 1 pins down the funding set but says nothing about its form. Under one more condition on the parameters, the funding set is an interval, so the firm is funded if and only if its dividend exceeds a threshold.

Assumption 2. $\phi > \alpha(\rho + \phi + \lambda - \mu)$.

Assumption 2 bounds the slope α of how much a firm needs to raise in each round. The ratio $\phi/(\rho + \phi + \lambda - \mu)$ is the equity value, per unit of dividend, when the equity holders expect no round to ever be funded, so that the first financing need kills the firm and the only source of value is a payout that arrives before it. The condition says that an additional unit of dividend raises the equity value by more than it raises the cost of the next round, even under this most pessimistic continuation conjecture. It ensures that $W(y) - \alpha y$ is strictly increasing, and it holds trivially when $\alpha = 0$.

Proposition 2 (Funding threshold). *Under Assumptions 1 and 2, in any symmetric Markov perfect equilibrium $W(y) - \alpha y$ is strictly increasing in y , so the funding set is an interval*

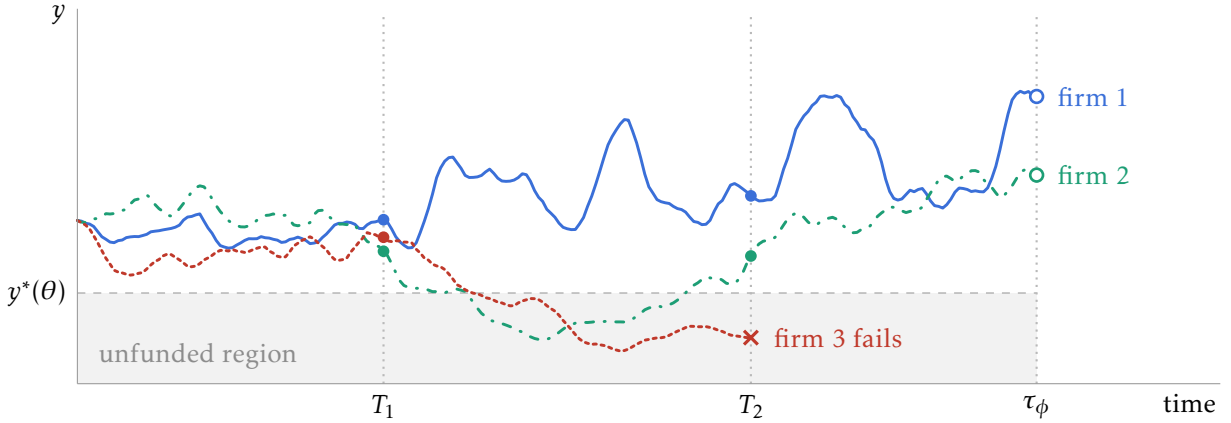
$$F = [y^*(\theta), \infty), \quad W(y^*(\theta)) = x(y^*(\theta))$$

The funding threshold $y^(\theta)$ is increasing in θ if $p > 0$, increasing in p if $\theta > 0$, increasing in α and in λ , and decreasing in σ if $p\theta < 1$.*

Proof: See Appendix A.2.

The threshold form of the equilibrium parallels the rollover threshold in He and Xiong (2012a), where each maturing creditor rolls over its debt if and only if the firm's fundamental exceeds a cutoff. Here each round is funded if and only if the dividend exceeds $y^*(\theta)$. The threshold is not a point of no return. A firm whose dividend falls below $y^*(\theta)$ does not fail at that moment. It fails only if a financing need arrives while the dividend is still below the threshold, so the dividend can cross the threshold and recover before the next round. Figure 2 illustrates this with three paths of the dividend that share the same dates of financing needs, T_1 and T_2 , and the same payout date τ_ϕ . Firm 1 stays above the threshold and is funded at both rounds. Firm 2 falls below the threshold after the first round, but no financing need arrives while it is there, and it has recovered by the time of the second round, which is funded, so it reaches the payout event like firm 1. Firm 3, the unlucky one, falls below the threshold after the first round and is still below it when the second financing need arrives. That round is not funded and the firm fails. What separates firms 2 and 3 is where the dividend stands when the financing need arrives.

Figure 2: Sample paths of the dividend and the funding threshold



The comparative statics follow from how each parameter moves the equity value, since the threshold is the dividend at which the equity value crosses the cost of the round. A financing need is always bad news for the equity holders. At an inside round either they pay the cost of the round themselves or the firm fails. At a market round either they are diluted or the firm fails. A higher λ makes financing needs more frequent, which lowers the equity value at every dividend and raises the threshold. A higher α works in the same direction. It raises the cost of every future round, which lowers the equity value at every dividend, and it raises the cost of the current round that the equity value must cover.

The effect of σ works through option value. Volatility matters because at every financing need the equity holders can walk away instead of paying $x(y)$. Their loss is capped at the equity they hold, while their gain when the dividend turns out high is not, so the payoff at a financing need is convex in the surplus $W - x$.

The wedge does not enter the funding condition itself. A round is funded if and only if $W(y) \geq x(y)$ whatever the value of θ . It moves the threshold only through the equity value. If $p > 0$, some future rounds will be market rounds, and at each of them the equity holders keep a fraction $1 - \theta$ of the surplus. A higher θ lowers what they keep at every future market round, which lowers the equity value at every dividend and raises the threshold. If $p = 0$ there are no market rounds and θ plays no role.

The effect of p is similar. If $\theta > 0$, a higher p increases the chance that a future round will be a market round, and at market rounds the equity holders keep a smaller fraction of the surplus than at inside rounds. This lowers the equity value and raises the threshold. If $\theta = 0$ a market round leaves the equity holders with the same surplus as an inside round, so p plays no role.

3 The Cost of Financing Frictions

This section defines the cost of the wedge. We first define the efficient benchmark for this economy as the funding policy that maximizes total surplus, the sum of the payoffs of the founder and of every investor. We then show that the market attains this benchmark when the wedge is removed. The market with $\theta = 0$ funds exactly the firms the efficient policy funds, and its threshold $y^*(0)$ is the efficient one. With the benchmark in hand, the cost of the wedge is the distance between the two thresholds. A firm whose dividend lies between $y^*(0)$ and $y^*(\theta)$ is worth funding and fails because of the wedge when a financing need arrives. The next subsection counts these firms. It introduces a population of firms and derives in closed form the share of failures the wedge causes and the share of failures that removing it would prevent, the two objects that the estimation later quantifies.

3.1. Efficient Benchmark

The efficient benchmark for this economy is the outcome that maximizes total surplus. The structure of the economy is that of Section 2. Financing needs arrive at rate λ , each is a market round with probability p and an inside round with probability $1 - p$, a market round can only be funded by a new investor, and a round costs $x(y)$ whoever funds it. A policy specifies two things. At each financing need, it decides whether the firm is funded or fails, and at each market round, it decides the stake s_i that the new investor receives. Total surplus is the sum of the expected discounted payoffs of the founder and of every new investor, with equal weights, conditional on the history h_t .

We keep the notation of Section 2.3. Let $d_j \in \{0, 1\}$ be the funding decision at round j , equal to one if the round is funded and to zero if the firm fails. The survival factor is

$$S(t, u) = \prod_{\substack{\text{all rounds } j \\ t < T_j \leq u}} d_j$$

Total surplus does not depend on the stakes. A stake is a transfer between players who have the same linear preferences and the same discount rate, so it redistributes the payoff without changing it, and what remains once the payoffs are added is the dividend, if the firm survives to the payout, net of the cost of every funded round, whoever puts the capital in.³ Taking

³Appendix A.3 gives the derivation.

expectations, the maximized total surplus solves the following problem:

$$V(h_t) = \max_{(d_j)_{j \geq 1}} \mathbb{E} \left[e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} S(t, \tau_\phi) - \sum_{\substack{\text{all rounds } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} x(y_{T_j}) S(t, T_j) \mid h_t \right] \quad (8)$$

where the maximum is over the funding decision d_j at every financing need after t , each of which may depend on the history at that date, and the stakes do not appear. Since the objective depends on the history only through the dividend path and the arrival dates, the efficient decision at a financing need depends on the history only through the current dividend, and V is a function of y . At a financing need the efficient policy compares the value of continuing net of the cost of the round, $V(y) - x(y)$, with the value of failure, which is zero, so it funds the firm if and only if $V(y) \geq x(y)$, and V satisfies the following HJB equation:

$$\rho V = \mu y V' + \frac{\sigma^2}{2} y^2 V'' + \phi(y - V) + \lambda(V - x(y))^+ - \lambda V \quad (9)$$

Compare this with the equilibrium HJB of the decentralized market, Eq. (7), which we restate here

$$\rho W = \mu y W' + \frac{\sigma^2}{2} y^2 W'' + \phi(y - W) + \lambda(1 - p\theta)(W - x(y))^+ - \lambda W$$

The two equations differ in one place. The HJB of the efficient benchmark is the market's with $p\theta = 0$. At $\theta = 0$, Assumption 1 becomes $\mu < \rho + \phi$, which implies it for every θ , and Assumption 2 does not involve θ . Under these two conditions the arguments of Propositions 1 and 2 apply to Eq. (9) without change, so V is its unique solution, $V - \alpha y$ is strictly increasing, and the efficient policy funds the firm if and only if the dividend exceeds a threshold, which we call y^{eff} . Writing $W(\cdot; 0)$ for the equity value of the market at $\theta = 0$, we have the following result.

Proposition 3 (Efficiency). *Under Assumption 2 and $\mu < \rho + \phi$, the maximized total surplus coincides with the equity value of the market without the wedge, and the efficient policy funds exactly the firms that the market without the wedge funds,*

$$V = W(\cdot; 0), \quad y^{\text{eff}} = y^*(0) \quad (10)$$

Proof. It follows from the discussion above. □

The market without the wedge funds exactly the firms that maximize total surplus. Its new investors receive the share x/W that their capital represents, which is a transfer and leaves total surplus unchanged, so the efficient allocation is implemented by pricing outside capital at its competitive level. We therefore refer to the market without the wedge as the

frictionless market from here on. With the benchmark in hand, the cost of the wedge is the distance between $y^*(\theta)$ and $y^*(0)$, which the next subsection measures.

3.2. Inefficient Failures

By Proposition 2, the funding threshold is increasing in the wedge whenever $p > 0$, so

$$\Delta(\theta) = y^*(\theta) - y^*(0) \geq 0 \quad (11)$$

with a strict inequality if $p\theta > 0$. A firm whose dividend lies in $[y^*(0), y^*(\theta))$ when a financing need arrives is worth funding, since the efficient benchmark funds it, but is not funded in the market. It fails because of the wedge, not because of its fundamentals. We will refer to these failures as inefficient.

We now turn to the main question of the paper: what fraction of startup failures does the wedge cause? We consider a continuum of firms, each following the model of Section 2 with the same parameters and independent shocks, and study the economy in its stationary state. It is convenient to work with the log dividend $u = \ln y$, which by Itô's lemma follows a Brownian motion with drift

$$du_t = m dt + \sigma dZ_t, \quad m = \mu - \frac{\sigma^2}{2}$$

and to write $u^* = \ln y^*(\theta)$ for the threshold in logs.

Firms enter the population at rate η , and the log dividend of an entrant is drawn from a density g with support in $[u^*, \infty)$. The support restriction reflects how firms enter the data: every firm in our sample raised a first round, so it was funded at entry, and a firm is funded only above the threshold. We place no other restriction on g , and the results below do not depend on it.

Let $f(u)$ be the stationary distribution of live firms over the log dividend, so that $f(u)du$ is the mass of live firms with log dividend in $[u, u + du)$. The stationary distribution satisfies the usual Kolmogorov forward equation

$$0 = \underbrace{-m f'(u) + \frac{\sigma^2}{2} f''(u)}_{\text{diffusion}} - \underbrace{\phi f(u)}_{\text{payout}} - \underbrace{\lambda \mathbb{1}_{\{u < u^*\}} f(u)}_{\text{failure}} + \underbrace{\eta g(u)}_{\text{entry}} \quad (12)$$

The first two terms are the net flow of firms into u from the drift and the volatility of the log dividend. The third is the flow of firms that leave at u because the payout arrives, at rate ϕ , wherever they are. The fourth is the flow that leaves because a financing need arrives while the dividend is below the threshold. Above the threshold a financing need is funded and the

firm stays where it is, so financing needs do not appear there. The last term is the flow of entrants at u .

Since we are interested in the fraction of failures that the wedge causes, and failures occur only below the threshold, we can focus on solving Eq. (12) there. Below u^* there is no entry, so the equation is homogeneous, and its solution depends on the entry distribution only through a constant that does not matter for the share of failures.

Proposition 4 (Inefficient failures). *In the stationary equilibrium, the share of failures that occur at dividends the efficient benchmark would fund is*

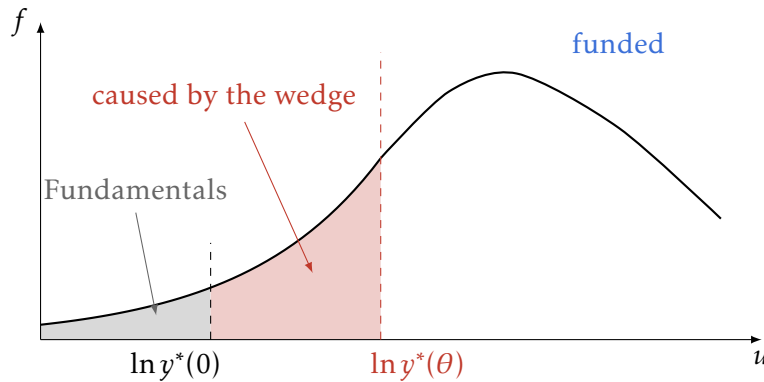
$$1 - \left(\frac{y^*(0)}{y^*(\theta)} \right)^\kappa, \quad \kappa = \frac{m + \sqrt{m^2 + 2\sigma^2(\phi + \lambda)}}{\sigma^2} > 0 \quad (13)$$

It does not depend on the entry rate η , on the entry distribution g , or on the slope α of the financing need.

Proof: See Appendix A.4.

Figure 3 shows the stationary density and the two groups of failures. Below $\ln y^*(\theta)$ every firm there fails when its next financing need arrives. The firms between $\ln y^*(0)$ and $\ln y^*(\theta)$ are worth funding and fail because of the wedge, and Eq. (13) is the share of the failure flow that comes from this region. The share depends on the parameters only through κ , which is determined by $(\mu, \sigma, \phi, \lambda)$, and through the ratio of the two thresholds.

Figure 3: Inefficient failures



Proposition 4 counts the firms that removing the wedge would save at the moment the wedge is removed. A firm in the region $[y^*(0), y^*(\theta))$ that is hit by a financing need is funded without the wedge, so that failure does not happen. Not all of these firms have a successful exit later, however. Some of them later drift below $y^*(0)$ and are hit by a financing need while there, and they fail even without the wedge, only later. We therefore ask a second question: of the firms that removing the wedge saves, what fraction survives to exit?

To answer it we compare the failure flow from the stationary distribution when the wedge is present with the new stationary distribution when the wedge is removed. We keep the entrants the same, the same rate η and the same density g , whose support remains $[\ln y^*(\theta), \infty)$, so that the comparison isolates the effect of the threshold. In the stationary equilibrium every entrant either fails or reaches the payout, so the failure flow is the entry rate times the probability that an entrant eventually fails, and this probability has a closed form on each side of the threshold.

Proposition 5 (Failures prevented). *In the stationary equilibrium, removing the wedge while keeping the entrants fixed reduces the failure flow by:*

$$1 - \left(\frac{y^*(0)}{y^*(\theta)} \right)^\zeta, \quad \zeta = \frac{m + \sqrt{m^2 + 2\sigma^2\phi}}{\sigma^2} > 0 \quad (14)$$

It does not depend on the entry rate η , on the entry distribution g , or on the slope α of the financing need.

Proof: See Appendix A.5.

The exponent ζ is the exponent κ of Proposition 4 with ϕ in place of $\phi + \lambda$, so $\zeta < \kappa$ and the share in Eq. (14) is smaller than the share in Eq. (13), as it should. The difference represents the firms that the wedge kills at one financing need and that would have failed at a later one anyway. The ratio of the two shares is the fraction of the firms saved at the moment of failure that go on to the payout. Notice that λ enters Eq. (14) only through the thresholds. The arrival rate of financing needs governs the probability that a firm fails once its dividend has reached the threshold, and that probability is the same under both thresholds, so it cancels in the ratio. What differs is how far an entrant's dividend has to drift before it reaches the threshold, and whether it gets there before the payout arrives, which is what ζ summarizes.

4 Data

This section describes the data and maps them to the model. We build a panel of venture-backed firms from PitchBook, reduce their deals to financing events that correspond to the model's rounds, reconstruct how each firm's history ends, and select the sample on which the model is estimated. The last part of the section states the choices that the mapping requires, in particular how a round is classified as inside or market and how failures are dated.

4.1 Source and Sample

Our data come from PitchBook, a commercial database of private-market transactions that records, for each deal, the company, the investors, the amount raised, the valuation and the

terms, and for each company its headquarters, its business status and its ownership status. We use the full export dated October 31, 2025, and everything measured in time, the observation windows, the failure dates and the sample cutoffs, is measured against that date.

We start from the 2.7 million deals in the export and keep venture capital deals that were completed and have a valid date. We then apply two filters at the company level. First, the company is headquartered in the United States. Second, its first venture capital deal is on or after January 1, 2005, so that every firm in the panel is observed from its first round and its round history is complete. Within these companies, the financing rounds are the primary financings, seed, angel, accelerator, early and later stage venture rounds, growth rounds and convertible notes. Debt, secondary transactions, grants and joint ventures are not financing rounds and are dropped. Acquisitions and buyouts are not financing rounds either, since the firm raises no equity in them, but they are how many firms leave the panel, and we use them below as exits, taken from the full list of deals. Finally, we require the firm's first priced round to be on or before October 31, 2018, seven years before the end of the data, so that every firm has enough time in the sample to exit or fail.⁴ The panel contains 41,722 firms whose first priced round falls between January 2005 and October 2018, observed through October 2025.

4.2. Financing Events

A round in the model is a single event at which the firm raises capital and sells equity. PitchBook records deals, which do not map one to one into rounds, for two reasons. First, a round often closes in several tranches, recorded as separate deals. We collapse the closes of the same stock series within ninety days of the first close into one financing event, summing the amounts and taking the valuation, the series and the price per share from the last close.⁵ Second, some financings are bridges rather than rounds. We classify an event as unpriced if it shows a convertible note or a SAFE and no evidence of pricing, no preferred or common stock, no series label, no price per share. Unpriced events are bridges to the next round, not rounds in the model's sense, so they stay in the panel but are not counted as rounds. Events before a firm's first priced round are dropped, so that the first round of every firm is a priced equity round.

For each priced round we build the following variables the estimation uses. The capital raised is the equity invested in the event, from PitchBook's total invested equity where reported and otherwise from the deal size net of any debt component. The stake sold is the percentage of the firm acquired by all the investors who invested in the round, from filings

⁴The cutoff is defined from the end of the data rather than by calendar year. Without it the most recent cohorts would be observed for too short a time to fail, and their presence would understate the failure rate. The choice of seven years for the cutoff follows [Ewens, Rhodes-Kropf and Strebulaev \(2016\)](#).

⁵The rule applies to 37 deals, 0.07% of the financing events in the estimation sample.

where reported and otherwise the ratio of the equity raised to the post-money valuation, and it is observed for 85% of the priced rounds in the estimation sample.⁶ The post-money valuation is observed for 84% of them. The investors in a round are the records with status new, follow-on or active investor, from which we count the investors and the new investors in each event. PitchBook also flags, for each investor record, whether that investor led the round.

The type of a round follows from its lead investor, for the reasons given in Section 5. A round is a market round if a new investor leads it and an inside round if every recorded lead is a returning investor. PitchBook records a lead for 38% of the priced rounds in the estimation sample, and rounds without a recorded lead are left untyped. Only the moments that involve the type of a round, and the probability p , use this field. A firm with an untyped round contributes that round to every other moment and stays in the sample.⁷ The first round of a firm is a market round in the model, and 99% of the first rounds with a recorded lead are led by a new investor in the data, so the moments that involve the type of a round, and p , are computed on rounds two and later⁸

4.3. Terminal Events

Each firm's history ends in one of three ways, an exit, a failure, or censoring at the end of the data. Exits take precedence over failures, that is, if a firm is first acquired and later fails, we consider it as having a successful exit. An exit is the firm's first initial public offering, merger or acquisition, or buyout dated on or after its entry into the panel, taken from the full list of deals. Firms with no such deal whose ownership status is acquired, publicly held or in registration are given an exit at the date of that status, without a valuation. The exit value is the post-money valuation of the exit deal. For an initial public offering it is the offer price times the shares outstanding after the issue. For an acquisition or a buyout it is the post-money valuation where reported and the deal size otherwise, and the two coincide where both exist.

A firm that has not exited is classified as failed by either of two rules. The first is PitchBook's own record, a business status of out of business or bankruptcy, or an ownership status of out of business. The second is a twelve-year rule, in the spirit of [Ewens, Rhodes-Kropf and Strebulaev \(2016\)](#), who treat a firm that raised its first round early enough and has not exited by the end of their data as a failed investment. A firm whose first priced round is at least twelve years before the end of the data, on or before October 31, 2013, and that has not exited,

⁶Where both measures exist they agree, and the ratio fallback covers a small selected set of rounds (3%).

⁷Requiring a recorded lead on every round would keep 13% of the firms and select short-lived firms with good outcomes. Rounds with a recorded lead are the larger financings, raising about three times as much and selling a stake 4 percentage points higher than untyped rounds of the same number, year and sector, so the wedge is measured on the better-documented rounds, and both types of round face the same reporting filter.

⁸We follow the same convention as in [Ewens, Rhodes-Kropf and Strebulaev \(2016\)](#).

is classified as failed. The failure date is the firm’s last recorded activity of any kind plus one year, capped at the end of the data for firms failed only by the twelve-year rule and active within the last year.⁹ A firm that has not exited or failed by the end of the data is censored at that date.

4.4. The Estimation Sample

The SMM part of the estimation simulates one model firm for each firm in the data, paired with that firm’s stake at entry and its observation window. It therefore requires the stake sold in the first round to be observed, which defines the estimation sample, or entry pool. Of the 41,722 firms in the panel, 16,521 have an observed first-round stake in $(0, 1]$. Six in ten firms drop at this step because PitchBook does not record the share sold in their first round, and since disclosure improves over time the sample leans toward the later cohorts within the 2005 to 2018 window. Table 1 reports summary statistics for the firms in the sample and for their priced rounds.

Table 2 summarizes the mapping between the model and the data that the rest of the paper uses. A financing need is a priced round. The stake is the share of the firm sold in the round. The payout is an exit, and the dividend at the payout is the exit valuation. A failure is a firm that fails as defined above, which in the model is a financing need that arrives while the dividend is below the threshold.

Table 2: Mapping the model to the data

Model	Data
Financing need, rate λ	Priced equity round
Inside round, market round	Round led by a returning investor, by a new investor
Stake s_i	Share of the firm sold in the round
Payout, rate ϕ	Initial public offering, acquisition or buyout
Dividend at the payout y_{τ_ϕ}	Valuation at the exit
Failure, $y < y^*$ at a financing need	Failure by status or by the twelve-year rule

5 Estimation

The model has nine parameters. Two describe the dividend, its drift μ and its volatility σ . Two describe the events in the firm’s life, the arrival rate λ of financing needs and the arrival rate

⁹In a later version of the paper we will use an additional rule to classify a firm as failed among those falling under the twelve-year rule. We have linked PitchBook and Revelio data, which gives a proxy for the employment level over time of startups based on LinkedIn data.

Table 1: Summary statistics of the estimation sample

	Mean	SD	p10	Median	p90	N
Panel A: Firms (16,521)						
<i>Sector (% of firms)</i>						
Information technology	46.3					16,521
Healthcare	21.4					16,521
Consumer products and services	15.6					16,521
Business products and services	11.7					16,521
Other	5.1					16,521
<i>Headquarters (% of firms)</i>						
California	41.1					16,521
New York	11.1					16,521
Massachusetts	8.6					16,521
<i>Terminal outcome by October 2025 (% of firms)</i>						
IPO	4.1					16,521
Acquired	35.0					16,521
Failed	29.7					16,521
Still active (censored)	31.2					16,521
<i>Firm characteristics</i>						
Year founded	2011	5	2005	2012	2016	16,223
Year of first round	2013	4	2007	2014	2018	16,521
Age at first round (years)	2.4	4.0	0.0	1.0	5.0	16,219
First-round raise (\$m)	4.91	15.48	0.40	2.15	10.00	16,465
First-round post-money (\$m)	18.0	140.7	2.3	8.7	31.6	16,355
First-round stake sold (%)	29.0	15.7	10.4	26.8	49.3	16,521
Priced rounds observed	3.1	2.0	1.0	3.0	6.0	16,521
Total equity raised (\$m)	54.7	265.4	1.0	11.3	118.1	16,482
Years observed	12.1	3.7	7.8	11.4	18.0	16,521
Panel B: Priced financing rounds (51,594)						
<i>Stage (% of rounds)</i>						
Seed	18.2					51,593
Early stage	38.6					51,593
Late stage	43.2					51,593
Lead investor recorded	37.8					51,594
Led by a new investor (market round)	82.4					19,521
Led by a new investor, rounds 2 and later	76.4					14,227
Down round	14.1					25,569
<i>Round characteristics</i>						
Round number	2.7	1.8	1.0	2.0	5.0	51,594
Number of investors	4.7	4.6	1.0	3.0	10.0	42,411
Years since previous round	1.81	1.34	0.56	1.49	3.38	35,073
Firm age at round (years)	4.9	4.4	1.0	4.0	10.0	50,749
Financing year	2016	5	2009	2016	2021	51,594
Equity raised (\$m)	18.3	104.4	0.6	5.0	38.1	49,268
Pre-money valuation (\$m)	139.8	1,284.5	3.0	15.6	175.0	43,457
Post-money valuation (\$m)	160.6	1,365.3	4.2	22.1	213.7	43,537
Stake sold (%)	24.5	14.7	7.7	22.4	43.9	43,756

ϕ of the payout. Two describe the financing need, its fixed component x_0 and its slope α . One describes the market, the probability p that a round is a market round, and one describes the frictions, the wedge θ . The last is the discount rate ρ . Of these, x_0 is a normalization. The model is homogeneous of degree one in the dividend and the financing need, so scaling both by a common factor scales the equity value and the threshold by the same factor, and setting $x_0 = 1$ fixes the unit in which dividends are measured.

We estimate the remaining eight parameters in four steps, ordered so that each step uses only what the previous ones deliver. First, two parameters need no estimation. The probability of a market round, p , is the share of rounds in the data that a new investor leads, and the discount rate ρ comes from the literature. Second, the two arrival rates, ϕ and λ , are estimated by maximum likelihood on the timing of rounds, payouts and failures, which the model ties to the two Poisson clocks. Third, the wedge θ is estimated from the difference between the stakes sold at market and at inside rounds, which the model ties to θ alone. Fourth, the drift and the volatility of the dividend, μ and σ , are estimated by the simulated method of moments, since the dividend is never observed and the moments that identify them depend on the whole path of the firm. The next three subsections describe the last three steps in turn.

Investors in the model are risk neutral and discount at rate ρ , so ρ is the expected return that investors require to hold venture equity between rounds. We take it from [Burguet, Maréchal and Mermoud \(2024\)](#), who revisit [Cochrane \(2005\)](#) on nearly 120,000 venture capital valuations from 2010 to 2022, correcting for the selection of firms into observed valuations. They estimate an annualized expected return on venture capital investments of around 38%, so we set $\rho = 0.32$.¹⁰

The probability of a market round is the share of priced rounds in the data at which a new lead investor enters, $p = 0.76$. We classify rounds by the lead investor rather than by any new investor for two reasons. The lead investor usually writes the largest check, so the round's capital comes mostly from it. The lead investor also sets the terms of the deal, the valuation and the stake, which the other participants accept. A round led by an investor already in the firm is priced by the owners, which is the model's inside round, whereas a round led by an outsider is priced by a newcomer who bargains with the owners, which is the model's analog of a market round.

¹⁰Their estimate is an arithmetic annual return, whereas ρ is continuously compounded, since the HJB equates ρ to the expected instantaneous return on the equity. The continuously compounded rate that delivers a 38% expected return over a year is $\rho = \ln(1.38) = 0.32$.

5.1. The Wedge

The model assumes that a market round is priced differently from an inside round, and the difference is the wedge. By Eq. (3), at a market round the new investor receives

$$s^M = \theta + (1 - \theta) \frac{x}{W}$$

whereas at an inside round the new shares are issued at the equity value and the share sold is $s^I = x/W$.¹¹ Hence $1 - s^M = (1 - \theta)(1 - x/W)$ and $1 - s^I = 1 - x/W$. For round k of firm i , with M_{ik} equal to one if the round is a market round and to zero otherwise, the model implies

$$\log(1 - s_{ik}) = \log(1 - \theta) M_{ik} + \log(1 - x_{ik}/W_{ik}) \quad (15)$$

Equation (15) implies the following regression

$$\log(1 - s_{ik}) = \beta M_{ik} + \beta_x \log x_{ik} + \gamma_{r(ik)} + \delta_{t(ik) \times c(i)} + \chi_i + \varepsilon_{ik} \quad (16)$$

which we estimate on the priced rounds two and later of the estimation sample with a recorded lead investor and an observed stake, restricted to the firms with at least two such rounds, so that every column, including the one with firm fixed effects, is estimated on the same rounds. The dependent variable is the log of the share of the firm that the owners retain after the round, one minus the stake sold. The coefficient of interest is β , and by Eq. (15) $\beta = \log(1 - \theta)$, so the wedge is recovered as $\theta = 1 - e^\beta$. The remaining terms stand in for the unobserved state $\log(1 - x_{ik}/W_{ik})$.¹² The state depends on the size of the round, so we include the log of the amount raised, $\log x_{ik}$, which holds the size of the round fixed and compares a market round to an inside round of the same size. $\gamma_{r(ik)}$ is a fixed effect for the round number $r(ik)$. The state also varies with market conditions, so $\delta_{t(ik) \times c(i)}$ is a fixed effect for the calendar year of the round $t(ik)$ interacted with the firm's sector $c(i)$. χ_i is a firm fixed effect, which compares the market and the inside rounds of the same firm. Table 3 reports the estimates without the size of the round, adding the fixed effects one column at a time, and Table 4 repeats every column with $\log x_{ik}$ included. We take as the baseline the estimate in column (4) of Table 4, $\theta = 0.03$, which includes every fixed effect and the size of the round. It is the most demanding specification. The firm fixed effect compares market and inside rounds of the same firm, which removes any persistent difference in the state of firms that raise more often from new

¹¹Admati and Pfleiderer (1994) show that an inside investor who holds a fixed fraction of the firm and finances the same fraction of every later round has no incentive to misprice those rounds, which is the sense in which an inside round is priced at the equity value.

¹²Note that W_{ik} is the total value of the firm at the time of round k , which is not the same as the post-money valuation in PitchBook. PitchBook's post-money valuation is calculated as x_{ik}/s_{ik} , which is only equal to the firm's value in inside rounds, or in the frictionless case.

investors, and holding the size of the round fixed removes the part of the gap that comes from market rounds being larger. It is also the lowest estimate across the specifications, which range from 0.030 to 0.050.

Table 3: The wedge from the pricing of market and inside rounds

	(1)	(2)	(3)	(4)
Market round	-0.0476*** (0.0046)	-0.0394*** (0.0044)	-0.0422*** (0.0042)	-0.0496*** (0.0046)
Implied $\theta = 1 - e^\beta$	0.0465 (0.0043)	0.0386 (0.0043)	0.0413 (0.0041)	0.0484 (0.0044)
Round-number FE		✓	✓	✓
Year $\times$ Sector FE			✓	✓
Company FE				✓
Observations	9,258	9,258	9,258	9,258
R ²	0.011	0.087	0.184	0.626

OLS estimates of $\log(1 - s_{ik}) = \beta M_{ik} + \gamma_{r(ik)} + \delta_{t(ik) \times c(i)} + \chi_i + \varepsilon_{ik}$ on priced rounds $k \geq 2$ of company i with a recorded lead investor and an observed stake s_{ik} . M_{ik} indicates a market round; $\gamma_{r(ik)}$, $\delta_{t(ik) \times c(i)}$ and χ_i are round-number, year $\times$ sector and company fixed effects. A market round is led by a new investor; an inside round by returning investors only. In the model $\beta = \log(1 - \theta)$, so $\theta = 1 - e^\beta$ (delta-method standard error). Fixed effects are added cumulatively across columns; the sample is identical across columns (companies with at least two typed later rounds). Standard errors clustered by company in parentheses.

Table 4: The wedge, holding the size of the round fixed

	(1)	(2)	(3)	(4)
Market round	-0.0513*** (0.0045)	-0.0375*** (0.0044)	-0.0407*** (0.0042)	-0.0301*** (0.0044)
log amount raised	0.0184*** (0.0018)	-0.0054** (0.0023)	-0.0045* (0.0023)	-0.0754*** (0.0040)
Implied $\theta = 1 - e^\beta$	0.0500 (0.0043)	0.0368 (0.0043)	0.0399 (0.0041)	0.0297 (0.0043)
Round-number FE		✓	✓	✓
Year $\times$ Sector FE			✓	✓
Company FE				✓
Observations	9,248	9,248	9,248	9,248
R ²	0.026	0.088	0.185	0.659

OLS estimates of $\log(1 - s_{ik}) = \beta M_{ik} + \beta_x \log x_{ik} + \gamma_{r(ik)} + \delta_{t(ik) \times c(i)} + \chi_i + \varepsilon_{ik}$ on priced rounds $k \geq 2$ of company i with a recorded lead investor, an observed stake s_{ik} , and a reported raise x_{ik} , companies with at least two such rounds; the sample is identical across columns. M_{ik} indicates a market round and x_{ik} is the equity raised in the round; $\gamma_{r(ik)}$, $\delta_{t(ik) \times c(i)}$ and χ_i are round-number, year $\times$ sector and company fixed effects. A market round is led by a new investor; an inside round by returning investors only. In the model $\beta = \log(1 - \theta)$, so $\theta = 1 - e^\beta$ (delta-method standard error). Fixed effects are added cumulatively across columns. Holding the raise fixed compares a market round to an inside round of the same size; since market rounds are larger and follow more growth, this is a lower bound on θ . Standard errors clustered by company in parentheses.

5.2. Maximum Likelihood

The two arrival rates govern the timing of everything the panel records. The dividend determines whether a round is funded, not when a financing need arrives, and it plays no role in when the payout arrives. We can therefore estimate ϕ and λ from dates alone.

For each firm i the panel records the dates of its K_i rounds beyond entry, the years T_i during which the firm is at risk, that is, from its first round to its exit, its failure or the end of the sample, and how its history ends, with an exit, a failure or censoring. Entry is conditioned on, not modeled, since every firm in the sample raised a first round. Let N_E be the number of firms observed exiting and N_F the number observed failing. The maximum likelihood

estimators are the two hazard rates

$$\hat{\phi} = \frac{N_E}{\sum_i T_i}, \quad \hat{\lambda} = \frac{\sum_i K_i + N_F}{\sum_i T_i} \quad (17)$$

The estimator of ϕ counts exits per year at risk. The estimator of λ counts financing needs per year at risk, and in the model a financing need has one of two outcomes. If the dividend is above the threshold the round is funded and the panel records a round, and if it is below, the round is not funded and the panel records a failure. Both are arrivals of the financing clock, so both are counted, and a firm's failure is its last financing need. Leaving failures out would treat a firm that fails as one whose last financing need never happened, and would bias $\hat{\lambda}$ downward.

The two estimators require no correction for the selection of firms into failure. The log-likelihood of a history is the sum of a term in ϕ , a term in λ , and a term that collects every condition on the path of the dividend, that funded rounds occurred above the threshold and failures below it. The last term depends on the dividend parameters and on the threshold but not on ϕ or λ , because the clocks are independent of the dividend, so it drops out of the first-order conditions for the arrival rates.¹³ The estimates of ϕ and λ are reported in Table 5.

5.3. Simulated Method of Moments

The remaining parameters are the drift and the volatility of the dividend, μ and σ . For now we set $\alpha = 0$, so that every round costs x_0 ¹⁴. The dividend is never observed, so neither parameter has a closed-form estimator, and we estimate them by the simulated method of moments. Given a candidate $\vartheta = (\mu, \sigma)$ and the parameters already in hand, we solve for the threshold, simulate a panel of firms with the same follow-up windows as the data, compute on the simulated panel the same two moments that we compute on the data, and search for the candidate that brings the two closest. The details of the simulation, the initial condition of the simulated firms, the treatment of censoring, and the computation of the moments are in Appendix B.

The two moments come from the firms that reach the payout, and they exploit a property of the model. The payout clock is independent of the dividend, so the date at which a firm's value is realized carries no information about that value, and conditional on the time T from entry to the payout, the log dividend at the payout is normal with mean $\ln y_0 + mT$ and variance $\sigma^2 T$, where $m = \mu - \sigma^2/2$. In a panel where every firm reaches the payout, a regression of the log exit value on T therefore recovers m as its slope, and a regression of its squared residuals

¹³This is also why μ and σ cannot be estimated from dates. They enter the likelihood only through the conditions on the dividend path, which the simulated method of moments handles next.

¹⁴In a later version of this paper, α will be jointly estimated with μ and σ in the SMM part.

on T recovers σ^2 . For each firm that reaches the payout, let lr_i be the log of the exit value relative to the capital raised at entry, and T_i the years from the first round to the payout. We run the two regressions

$$lr_i = a + b T_i + e_i, \quad e_i^2 = c + d T_i + u_i \quad (18)$$

and use the two slopes.

- (i) The slope b is the average growth rate of the log dividend among the firms that reach the payout, which under the geometric Brownian motion is $m = \mu - \sigma^2/2$, up to the selection of the firms that survive to the payout. Together with the volatility from the next moment, it pins down the drift μ .
- (ii) The slope d is the rate at which the dispersion of the log dividend grows with time, which under the geometric Brownian motion is σ^2 . It pins down the volatility.

Two moments identify two parameters, so the model is exactly identified.

In the data, only the firms that reach the payout have an exit value, and reaching the payout requires surviving every financing need along the way, which selects the paths that stayed above the threshold. The two slopes are therefore not m and σ^2 themselves, and we do not invert them. The simulated panel is subject to the same selection, since a simulated firm fails when its dividend is below the threshold at a financing need and only the simulated firms that reach the payout enter the regressions, so the simulated and the data slopes are computed on samples selected in the same way. The selection is what makes the slopes depend on the threshold, and through it on the wedge and the arrival rates, as well as on m and σ .

We estimate ϑ by minimizing the sum of squared symmetric percentage deviations between the simulated and the data moments,

$$\hat{\vartheta} = \arg \min_{\vartheta} \sum_{j=1}^2 \left[\frac{m_j(\vartheta) - \hat{m}_j}{(|m_j(\vartheta)| + |\hat{m}_j|)/2} \right]^2 \quad (19)$$

where $m_j(\vartheta)$ is the j -th moment on the simulated panel and $\hat{m}_j$ its counterpart in the data. The deviations are scale free, so the two slopes, which differ in magnitude, contribute equally. With two moments and two parameters the weighting does not matter at the solution, where both moments are matched.

Table 5 shows the parameters, their values and where each comes from.

Table 5: Parameters

Parameter	Value	Source
x_0	Financing need, fixed component	1 Normalization
α	Financing need, slope	0 Assumption
ρ	Discount rate	0.32 Burguet, Maréchal and Mermoud (2024)
p	Probability of a market round	0.76 Share of rounds led by a new investor
θ	Wedge	0.03 Table 4, column (4)
ϕ	Arrival rate of the payout	0.051 Maximum likelihood, Eq. (17)
λ	Arrival rate of financing needs	0.315 Maximum likelihood, Eq. (17)
μ	Drift of the dividend	0.275 SMM
σ	Volatility of the dividend	0.522 SMM

6 Results

With the parameters in hand, the cost of the wedge follows from Section 3 in closed form. Table 6 reports the two thresholds, the two exponents, and the two shares of Propositions 4 and 5 at the estimated parameters.

Table 6: The cost of the wedge at the estimated parameters

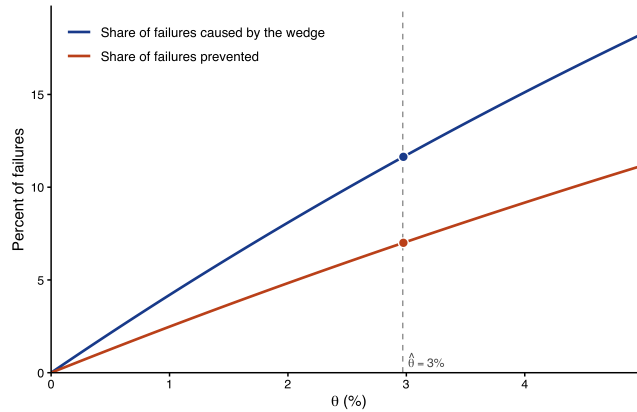
Quantity	Value
Funding threshold $y^*(\theta)$	3.44
Efficient threshold $y^*(0)$	3.26
Ratio of thresholds $y^*(0)/y^*(\theta)$	0.95
Density exponent κ	2.23
Survival exponent ζ	1.31
Share of failures caused by the wedge (%)	11.62
Share of failures prevented (%)	6.99

The funding threshold with the wedge is $y^*(\theta) = 3.44$, in units of the financing need. A firm is funded at a round only if its dividend is at least 3.44 times the cost of the round, which is what it takes for the equity to be worth the capital it needs, given that the payout is far away and that the firm will face several more rounds before it. Without the wedge the threshold is $y^*(0) = 3.26$. A wedge of 3% of the surplus at each market round raises the threshold by 5.2%, because the wedge is paid at every market round between now and the payout, and the equity value today already prices all of them.

The two shares translate the distance between the thresholds into failures. By Proposition 4, 11.6% of the failures in the stationary equilibrium occur at firms that the efficient benchmark funds, those whose dividend lies between $y^*(0)$ and $y^*(\theta)$ when a financing need arrives. The region within 5.2% of the threshold with the wedge accounts for more than twice that share of the failures because the density of firms below the threshold rises steeply toward it. By Proposition 5, removing the wedge lowers the failure flow by 7.0%. The ratio of the two shares is 0.60: of the firms that the wedge kills and that would be funded without it, 60% make it to the payout, and the remaining 40% drift below $y^*(0)$ and fail at a later financing need even without the wedge.

Figure 4 traces the two shares as the wedge varies from zero to 5%, holding every other parameter at its estimated value¹⁵. This figure shows, for each wedge in the range, what share of the failures occur at firms that the efficient benchmark funds, and what share of the failure flow would be prevented by removing the wedge. Both shares rise with the wedge, and nearly in proportion to it over this range. The estimates of θ in Tables 3 and 4 run from 3% to 5%, and the figure translates that range into a range for the cost. At a wedge of 5%, the upper end of the estimates, about 18% of the failures would occur at firms worth funding and removing the wedge would prevent about 11% of the failure flow, against 11.6% and 7.0% at the baseline estimate of 3%. The figures in Section 6 are therefore at the low end of what the estimates of the wedge imply.

Figure 4: The cost of the wedge as a function of the wedge



The cost of the wedge also depends on the other parameters, and Appendix C traces the two shares as each of them varies around its estimate, one at a time, holding the rest fixed. Both shares rise with the arrival rate of financing needs and with the probability of a market round, and fall with the arrival rate of the payout and with the discount rate. The discount

¹⁵The estimates of μ and σ are not updated along the way, that is, we keep the estimates of μ and σ from the initial SMM estimation with $\theta = 3\%$. For the range in the figure, we do not expect the estimates of μ and σ to change much.

rate is the parameter to which the cost is most sensitive.

7 Conclusion

Startups raise capital in a sequence of rounds, and every round carries a risk of failure. This paper asks how much of startup failure is due to the terms on which the next round is priced rather than to the prospects of the firm. We answer it with a model of staged equity financing in which a firm with a stochastic dividend must repeatedly raise capital, either from its current owners or from a new investor who bargains for a stake, and in which financing frictions are summarized by a wedge between the price of outside capital and its competitive level.

The model delivers two results. First, because equity is repriced at every round, a new investor is compensated for the risk of future rounds through the stake it receives rather than exposed to it at a fixed price. Repricing substitutes for coordination, the equilibrium is unique, and staged equity does not admit the self-fulfilling failures of debt. Second, a firm fails only when a financing need arrives while its dividend is below a threshold, and the wedge raises that threshold. The market without the wedge funds exactly the firms that maximize total surplus, so the firms between the two thresholds fail because of the wedge alone. The share of failures that occur at such firms, and the share of the failure flow that removing the wedge would prevent, are presented in closed form and do not depend on how firms enter or on the slope of the financing need.

We estimate the model on a panel of venture-backed firms from PitchBook. The wedge is identified by the difference between the stakes sold at rounds led by new investors and at rounds led by returning ones. At the estimated parameters the wedge raises the funding threshold by 5.2%, 11.6% of the failures in the stationary equilibrium occur at firms that are worth funding, and removing the wedge would lower the failure flow by 7.0%, so six in ten of the firms it saves at the moment of failure go on to have a successful exit. A wedge that is small at every round has a cost that compounds over the rounds a firm expects to face, which is why financing frictions matter for firms that have to return to the market again.

Two extensions are worth pursuing. The type of a round is exogenous in the model, whereas in the data firms may seek out new investors after good news and turn to their owners after bad news, and a model of that choice would sharpen the identification of the wedge. The financing need is constant in the estimation, whereas rounds grow with valuations in the data, and letting the need scale with the dividend would bring the later-round stakes of the model closer to those observed.

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A Omitted Proofs

A.1. Proof of Proposition 1

Proof. Fix a symmetric Markov perfect equilibrium. By Definition 1 its funding sets coincide, $F = F^0 = \{y : W(y) \geq x(y)\}$, where $W = W^{F,F}$ is its equity value. We show that W is the fixed

point of a contraction, so that any two equilibria have the same equity value and therefore the same funding set.

Let τ_λ be the arrival time of the first financing need, $\tau = \min\{\tau_\phi, \tau_\lambda\}$ the time of the first event, and $\mathbb{E}_y$ the expectation conditional on $y_0 = y$. At a financing need with dividend y , the equity holders keep all of the surplus at an inside round and a fraction $1 - \theta$ of it at a market round, and the round is funded if and only if $y \in F$, so their expected payoff is

$$\psi(y) = (1 - p)(W(y) - x(y)) \mathbb{1}_{\{y \in F\}} + p(1 - \theta)(W(y) - x(y)) \mathbb{1}_{\{y \in F\}} = (1 - p\theta)(W(y) - x(y))^+$$

where the last equality uses $F = \{y : W(y) \geq x(y)\}$. We can write $W(y)$ as the expected discounted payoff at the first event, which is either a payout or a financing need:

$$W(y) = \mathbb{E}_y \left[e^{-\rho\tau} \left(y_\tau \mathbb{1}_{\{\tau_\phi < \tau_\lambda\}} + \psi(y_\tau) \mathbb{1}_{\{\tau_\lambda < \tau_\phi\}} \right) \right]$$

The clocks τ_ϕ and τ_λ are independent exponentials with rates ϕ and λ , independent of the dividend. Conditioning on $\tau_\phi = s$ and $\tau_\lambda = t$ and using the law of iterated expectations,

$$\begin{aligned} W(y) &= \int_0^\infty \int_0^\infty \phi e^{-\phi s} \lambda e^{-\lambda t} \left[e^{-\rho s} \mathbb{E}_y[y_s] \mathbb{1}_{\{s < t\}} + e^{-\rho t} \mathbb{E}_y[\psi(y_t)] \mathbb{1}_{\{t < s\}} \right] ds dt \\ &= \int_0^\infty \phi e^{-(\phi+\rho)s} \mathbb{E}_y[y_s] \left(\lambda \int_s^\infty e^{-\lambda t} dt \right) ds + \int_0^\infty \lambda e^{-(\lambda+\rho)t} \mathbb{E}_y[\psi(y_t)] \left(\int_t^\infty \phi e^{-\phi s} ds \right) dt \\ &= \int_0^\infty e^{-(\lambda+\rho+\phi)t} \mathbb{E}_y[\phi y_t + \lambda \psi(y_t)] dt \end{aligned}$$

which yields the fixed point equation:

$$W(y) = \mathbb{E}_y \left[\int_0^\infty e^{-(\lambda+\rho+\phi)t} \left[\phi y_t + \lambda (1 - p\theta)(W(y_t) - x(y_t))^+ \right] dt \right] \quad (\text{A.1})$$

Let $\mathcal{B}$ be the space of measurable functions $V : (0, \infty) \rightarrow \mathbb{R}$ with the following norm:

$$\|V\| = \sup_{y>0} \frac{|V(y)|}{1 + y}$$

For $V \in \mathcal{B}$ define:

$$(TV)(y) = \mathbb{E}_y \left[\int_0^\infty e^{-(\lambda+\rho+\phi)t} \left[\phi y_t + \lambda (1 - p\theta)(V(y_t) - x(y_t))^+ \right] dt \right]$$

Since $|V(y_t)| \leq \|V\|(1 + y_t)$ and $\mathbb{E}_y[y_t] = ye^{\mu t}$ with $\mu < \lambda + \rho + \phi$, TV has finite norm, so T maps $\mathcal{B}$ into itself, and Eq. (A.1) says that the equilibrium equity value is a fixed point of T .

For $V_1, V_2 \in \mathcal{B}$, the payout terms cancel and

$$(TV_1)(y) - (TV_2)(y) = \lambda(1 - p\theta) \mathbb{E}_y \left[\int_0^\infty e^{-(\lambda+\rho+\phi)t} \left[(V_1(y_t) - x(y_t))^+ - (V_2(y_t) - x(y_t))^+ \right] dt \right]$$

For a, b, x we have: $|(a - x)^+ - (b - x)^+| \leq |a - b|$. Taking absolute values and using $|V_1(y_t) - V_2(y_t)| \leq \|V_1 - V_2\|(1 + y_t)$,

$$\begin{aligned} |(TV_1)(y) - (TV_2)(y)| &\leq \lambda(1 - p\theta) \|V_1 - V_2\| \int_0^\infty e^{-(\lambda+\rho+\phi)t} (1 + ye^{\mu t}) dt \\ &= \lambda(1 - p\theta) \|V_1 - V_2\| \left(\frac{1}{\lambda + \rho + \phi} + \frac{y}{\lambda + \rho + \phi - \mu} \right) \\ &\leq \frac{\lambda(1 - p\theta)}{\min\{\lambda + \rho + \phi, \lambda + \rho + \phi - \mu\}} \|V_1 - V_2\| (1 + y) \end{aligned}$$

Dividing by $1 + y$ and taking the supremum over $y > 0$,

$$\|TV_1 - TV_2\| \leq q \|V_1 - V_2\|, \quad q = \frac{\lambda(1 - p\theta)}{\min\{\lambda + \rho + \phi, \lambda + \rho + \phi - \mu\}}$$

If $\mu \geq 0$ the minimum is $\lambda + \rho + \phi - \mu$, and $q < 1$ if and only if $\lambda(1 - p\theta) < \lambda + \rho + \phi - \mu$, that is, $\mu < \rho + \phi + \lambda p\theta$. If $\mu < 0$ the minimum is $\lambda + \rho + \phi$, and $q < 1$ if and only if $0 < \rho + \phi + \lambda p\theta$, which holds trivially. In both cases $q < 1$ under Assumption 1, so T is a contraction on $\mathcal{B}$ and has at most one fixed point.

Any two symmetric Markov perfect equilibria therefore have the same equity value W , hence the same funding set $F = \{y : W(y) \geq x(y)\}$ and, by Eq. (3), the same stake at every market round, which is the statement of Proposition 1. $\square$

A.2. Proof of Proposition 2

Proof. Throughout, W is the equilibrium equity value, which solves Eq. (7), and we write

$$R = \rho + \phi + \lambda p\theta - \mu, \quad S = \rho + \phi + \lambda - \mu$$

Both are positive under Assumption 1, and $S - R = \lambda(1 - p\theta)$. We first conjecture that $W(y) - x(y)$ is strictly increasing, solve Eq. (7) on each side of the threshold this conjecture implies, and pin down the threshold from the value of the equity there. The remaining steps verify the conjecture and establish the comparative statics.

The equity value on each side of the threshold. Suppose that $W(y) - x(y)$ is strictly increasing. Then $W(y) \geq x(y)$ if and only if $y \geq y^*$, where y^* solves $W(y^*) = x(y^*)$, and Eq. (7) is a

linear second-order differential equation on each side of y^* .

In the unfunded region, $y < y^*$, the surplus term vanishes and Eq. (7) becomes

$$(\rho + \phi + \lambda) W = \mu y W' + \frac{\sigma^2}{2} y^2 W'' + \phi y$$

A linear particular solution $W = my$ requires $(\rho + \phi + \lambda)m = \mu m + \phi$, that is, $m_U = \phi/S$. The homogeneous equation has solutions y^γ , with γ a root of the characteristic equation

$$\frac{\sigma^2}{2} \gamma^2 + \left(\mu - \frac{\sigma^2}{2} \right) \gamma - (\rho + \phi + \lambda) = 0$$

The product of the roots is $-2(\rho + \phi + \lambda)/\sigma^2 < 0$, so one root is negative and the other positive, and since the left side equals $\mu - (\rho + \phi + \lambda) < 0$ at $\gamma = 1$, the positive root exceeds one. The negative root is ruled out, since if $\gamma < 0$, y^γ diverges as $y \rightarrow 0$, whereas the equity value is bounded near zero. Hence

$$W(y) = m_U y + B y^\gamma, \quad \gamma = \frac{-\left(\mu - \frac{\sigma^2}{2}\right) + \sqrt{\left(\mu - \frac{\sigma^2}{2}\right)^2 + 2\sigma^2(\rho + \phi + \lambda)}}{\sigma^2} > 1$$

for some constant B .

In the funded region, $y \geq y^*$, the surplus is $W - \alpha y - x_0$ and Eq. (7) becomes

$$(\rho + \phi + \lambda p\theta) W = \mu y W' + \frac{\sigma^2}{2} y^2 W'' + (\phi - \alpha\lambda(1 - p\theta))y - \lambda(1 - p\theta)x_0$$

A particular solution of the form $W = m_F y - c$ requires, matching the coefficients of y and the constants,

$$m_F = \frac{\phi - \alpha\lambda(1 - p\theta)}{R}, \quad c = \frac{\lambda(1 - p\theta)x_0}{\rho + \phi + \lambda p\theta}$$

The slope m_F is positive under Assumptions 1 and 2, since $\phi > \alpha S \geq \alpha\lambda(1 - p\theta)$, where the first inequality is Assumption 2 and the second is $\alpha R \geq 0$. The homogeneous equation has solutions y^β , with β a root of

$$\frac{\sigma^2}{2} \beta^2 + \left(\mu - \frac{\sigma^2}{2} \right) \beta - (\rho + \phi + \lambda p\theta) = 0$$

As before, one root is negative and the other positive, and since the left side equals $\mu - (\rho + \phi + \lambda p\theta) < 0$ at $\beta = 1$ by Assumption 1, the positive root exceeds one. The positive root is ruled out since if $\beta > 1$, y^β grows faster than y , whereas the equity value grows at most

linearly. Hence

$$W(y) = m_F y - c + A y^\beta, \quad \beta = \frac{-\left(\mu - \frac{\sigma^2}{2}\right) - \sqrt{\left(\mu - \frac{\sigma^2}{2}\right)^2 + 2\sigma^2(\rho + \phi + \lambda p\theta)}}{\sigma^2} < 0$$

for some constant A . Collecting the two regions,

$$W(y) = \begin{cases} m_F y - c + A y^\beta & y \geq y^* \\ m_U y + B y^\gamma & y < y^* \end{cases} \quad (\text{A.2})$$

Neither β nor γ depends on α . Of the two, only β depends on θ , through $\lambda p\theta$.

The threshold. The constants A and B and the threshold y^* are pinned down by three conditions: W is continuous at y^* , W' is continuous at y^* , and the equity value equals the cost of the round at the threshold, $W(y^*) = \alpha y^* + x_0$.

The first two conditions give, after multiplying the second condition by y^* ,

$$\begin{aligned} m_F y^* - c + A (y^*)^\beta &= m_U y^* + B (y^*)^\gamma \\ m_F y^* + \beta A (y^*)^\beta &= m_U y^* + \gamma B (y^*)^\gamma \end{aligned}$$

Writing $a = A (y^*)^\beta$ and $b = B (y^*)^\gamma$ for the two homogeneous terms at the threshold, this is a linear system in (a, b) ,

$$a - b = (m_U - m_F) y^* + c, \quad \beta a - \gamma b = (m_U - m_F) y^*$$

with solution

$$a = \frac{(\gamma - 1)(m_U - m_F) y^* + \gamma c}{\gamma - \beta}, \quad b = \frac{(\beta - 1)(m_U - m_F) y^* + \beta c}{\gamma - \beta}$$

The third condition is the definition of the threshold, $W(y^*) = \alpha y^* + x_0$. Evaluating W at y^* from the unfunded side of Eq. (A.2) and substituting b ,

$$W(y^*) = m_U y^* + b = \frac{(\gamma - 1)m_U + (1 - \beta)m_F}{\gamma - \beta} y^* + \frac{\beta c}{\gamma - \beta}$$

Setting this equal to $\alpha y^* + x_0$ and solving for y^* gives

$$y^* = \frac{x_0(\gamma - \beta) - \beta c}{(\gamma - 1)m_U + (1 - \beta)m_F - \alpha(\gamma - \beta)} \quad (\text{A.3})$$

The numerator is positive, since $\gamma > 1$, $\beta < 0$ and $c > 0$. We can simplify the denominator further. Note that:

$$(\gamma - 1)m_U + (1 - \beta)m_F - \alpha(\gamma - \beta) = \phi \left[\frac{\gamma - 1}{S} + \frac{1 - \beta}{R} \right] - \alpha \left[\gamma - \beta + \frac{(1 - \beta)\lambda(1 - p\theta)}{R} \right]$$

The two brackets are proportional. Using $\lambda(1 - p\theta) = S - R$,

$$\gamma - \beta + \frac{(1 - \beta)(S - R)}{R} = \frac{(\gamma - 1)R + (1 - \beta)S}{R} = \frac{S}{\phi} \left[\frac{\gamma - 1}{S} + \frac{1 - \beta}{R} \right]$$

so the denominator of Eq. (A.3) equals $\phi \left[\frac{\gamma - 1}{S} + \frac{1 - \beta}{R} \right] (1 - \alpha S / \phi)$, and

$$y^*(\theta) = \frac{\phi}{\phi - \alpha S} \tilde{y}^*(\theta), \quad \tilde{y}^*(\theta) = \frac{x_0}{\phi} \cdot \frac{\gamma - \beta \frac{\rho + \phi + \lambda}{\rho + \phi + \lambda p \theta}}{\frac{\gamma - 1}{S} + \frac{1 - \beta}{R}} \quad (\text{A.4})$$

where $\tilde{y}^*(\theta)$ is the threshold when $\alpha = 0$. The bracket in the denominator of $\tilde{y}^*$ is positive, so $\tilde{y}^*(\theta) > 0$ when $x_0 > 0$, and Assumption 2 is exactly the condition $\phi - \alpha S > 0$ that makes $y^*(\theta)$ positive and finite. The factor $\phi / (\phi - \alpha S)$ does not depend on θ , so the slope of the financing need scales the threshold by the same factor for every value of the wedge.

$W(y) - \alpha y$ is strictly increasing below the threshold. We now verify the conjecture, starting with the unfunded region. By Eq. (A.2), for $y < y^*$,

$$W(y) - \alpha y - x_0 = (m_U - \alpha)y + B y^\gamma - x_0$$

The first term is strictly increasing, since $m_U - \alpha = (\phi - \alpha S) / S > 0$ by Assumption 2, and the second is nondecreasing if $B \geq 0$, since $\gamma > 1$. It remains to sign B . Evaluating the unfunded branch at the threshold, where $W(y^*) = \alpha y^* + x_0$,

$$B(y^*)^\gamma = x_0 - (m_U - \alpha)y^* = x_0 - m_U \tilde{y}^*(\theta)$$

where the second equality uses Eq. (A.4) and $m_U - \alpha = (\phi - \alpha S) / S$, so the right side does not depend on α . Substituting $\tilde{y}^*$ from Eq. (A.4) and using $m_U = \phi / S$,

$$x_0 - m_U \tilde{y}^*(\theta) = \frac{x_0}{\gamma - 1 + (1 - \beta) \frac{S}{R}} \left[-1 + (1 - \beta) \frac{S}{R} + \beta \frac{\rho + \phi + \lambda}{\rho + \phi + \lambda p \theta} \right]$$

Writing $S/R = 1 + \lambda(1 - p\theta)/R$ and $(\rho + \phi + \lambda)/(\rho + \phi + \lambda p\theta) = 1 + \lambda(1 - p\theta)/(\rho + \phi + \lambda p\theta)$, the constant terms in the bracket cancel and it equals

$$\lambda(1 - p\theta) \left[\frac{1 - \beta}{R} + \frac{\beta}{\rho + \phi + \lambda p\theta} \right] = \lambda(1 - p\theta) \frac{\rho + \phi + \lambda p\theta - \mu\beta}{R(\rho + \phi + \lambda p\theta)}$$

The characteristic equation for β can be rearranged as $\frac{\sigma^2}{2}\beta(\beta - 1) = \rho + \phi + \lambda p\theta - \mu\beta$, and the left side is positive because $\beta < 0$. Every factor is therefore positive, and

$$B(y^*)^\gamma = \frac{x_0 \lambda(1 - p\theta)}{\gamma - 1 + (1 - \beta) \frac{S}{R}} \cdot \frac{\rho + \phi + \lambda p\theta - \mu\beta}{R(\rho + \phi + \lambda p\theta)} \geq 0 \quad (\text{A.5})$$

with equality only if $p\theta = 1$. This gives us that $B \geq 0$. Hence $W(y) - \alpha y$ is strictly increasing on $(0, y^*)$, and since it equals x_0 at y^* , $W(y) < x(y)$ for every $y < y^*$, as conjectured.

$W(y) - \alpha y$ is strictly increasing above the threshold. By Eq. (A.2), for $y > y^*$,

$$\frac{d}{dy} [W(y) - \alpha y] = m_F - \alpha + A\beta y^{\beta-1}, \quad m_F - \alpha = \frac{\phi - \alpha \lambda(1 - p\theta) - \alpha R}{R} = \frac{\phi - \alpha S}{R} > 0$$

where the inequality is Assumption 2 again. If $A \leq 0$, the second term is nonnegative because $\beta < 0$, and the derivative is positive. If $A > 0$, the second term is negative, but

$$\frac{d^2}{dy^2} [W(y) - \alpha y] = A\beta(\beta - 1)y^{\beta-2} > 0$$

since $\beta < 0$ and $\beta - 1 < 0$, so the derivative is increasing on (y^*, ∞) . Its value as $y \downarrow y^*$ equals its value as $y \uparrow y^*$, because W' is continuous at y^* , and the latter is $m_U - \alpha + \gamma B(y^*)^{\gamma-1} > 0$ by the previous paragraph. Hence the derivative is positive on (y^*, ∞) in this case as well.

Together with the result below the threshold, $W(y) - \alpha y$ is strictly increasing on $(0, \infty)$, so

$$\{y : W(y) \geq x(y)\} = [y^*, \infty)$$

and the conjecture holds. Since T has exactly one fixed point, every symmetric Markov perfect equilibrium has this equity value and this funding set, which proves the first part of Proposition 2.

Comparative statics: α . None of β , γ , R and S depends on α , so $\tilde{y}^*(\theta)$ in Eq. (A.4) does not depend on α either, and

$$\frac{\partial y^*(\theta)}{\partial \alpha} = \tilde{y}^*(\theta) \frac{\phi S}{(\phi - \alpha S)^2} > 0$$

on the range $\alpha < \phi/S$ allowed by Assumption 2. Thus, the threshold is increasing in α .

Simplifying $\tilde{y}^*$. The remaining parameters enter Eq. (A.4) through $\tilde{y}^*(\theta)$ and, in the case of λ , through S . It is convenient to write

$$v = \frac{\sigma^2}{2}, \quad Q = \rho + \phi + \lambda = S + \mu, \quad q = \rho + \phi + \lambda p\theta = R + \mu$$

so that the two characteristic equations are $v\gamma^2 + (\mu - v)\gamma - Q = 0$ and $v\beta^2 + (\mu - v)\beta - q = 0$, and $\tilde{y}^*(\theta) = \frac{x_0}{\phi} \left(\gamma - \beta \frac{Q}{q} \right) / \left(\frac{\gamma-1}{Q-\mu} + \frac{1-\beta}{q-\mu} \right)$.

We have the following identities for any root z of $vz^2 + (\mu - v)z - r = 0$,

$$r - \mu = vz^2 + (\mu - v)z - \mu = (z - 1)(vz + \mu), \quad r = z[v(z - 1) + \mu]$$

Applying the first identity to γ and β , the denominator of $\tilde{y}^*$ becomes

$$\frac{\gamma - 1}{Q - \mu} + \frac{1 - \beta}{q - \mu} = \frac{1}{\gamma v + \mu} - \frac{1}{\beta v + \mu} = \frac{(\beta - \gamma)v}{(\gamma v + \mu)(\beta v + \mu)}$$

and applying the second to Q and q , the numerator becomes

$$\gamma - \beta \frac{Q}{q} = \gamma \left[1 - \frac{v(\gamma - 1) + \mu}{v(\beta - 1) + \mu} \right] = \frac{\gamma v(\beta - \gamma)}{v(\beta - 1) + \mu}$$

Taking the ratio, the factor $(\beta - \gamma)v$ cancels and

$$\tilde{y}^*(\theta) = \frac{x_0}{\phi} \gamma \frac{(\mu + \gamma v)(\mu + \beta v)}{\mu + v(\beta - 1)} \quad (\text{A.6})$$

First, $\mu + \gamma v > 0$, since $\gamma(\mu + \gamma v) = Q + \gamma v > 0$ and $\gamma > 0$. Second, $\mu + \beta v < 0$, since $R = q - \mu = (\beta - 1)(\beta v + \mu) > 0$ and $\beta - 1 < 0$. Third, $\mu + v(\beta - 1) = (\mu + \beta v) - v < 0$. The last two factors are negative, so $\tilde{y}^*(\theta) > 0$, as found above.

Comparative statics: θ and p . The wedge and the probability of a market round enter Eq. (A.6) only through β , and β depends on them only through $q = \rho + \phi + \lambda p\theta$. Neither affects γ , v or the factor $\phi/(\phi - \alpha S)$ in Eq. (A.4). Differentiating the factor that contains β ,

$$\frac{d}{d\beta} \left[\frac{\mu + \beta v}{\mu + v(\beta - 1)} \right] = \frac{v[\mu + v(\beta - 1)] - v(\mu + \beta v)}{[\mu + v(\beta - 1)]^2} = \frac{-v^2}{[\mu + v(\beta - 1)]^2} < 0$$

so the threshold is decreasing in β . Differentiating the characteristic equation of β implicitly with respect to q ,

$$\frac{\partial \beta}{\partial q} = \frac{1}{2v\beta + \mu - v} < 0$$

where the denominator is the slope of the convex quadratic $vz^2 + (\mu - v)z - q$ at its smaller root, which is negative. By the chain rule,

$$\frac{\partial y^*(\theta)}{\partial \theta} = \frac{\partial y^*(\theta)}{\partial \beta} \frac{\partial \beta}{\partial q} \lambda p, \quad \frac{\partial y^*(\theta)}{\partial p} = \frac{\partial y^*(\theta)}{\partial \beta} \frac{\partial \beta}{\partial q} \lambda \theta$$

The product of the first two factors is positive, so the threshold is strictly increasing in θ if $p > 0$ and strictly increasing in p if $\theta > 0$.

Comparative statics: λ . The arrival rate of financing needs enters $y^*(\theta)$ in three ways: through Q in the characteristic equation of γ , through q in the characteristic equation of β , and through S in the factor $\phi/(\phi - \alpha S)$ of Eq. (A.4). Write Eq. (A.6) as

$$\tilde{y}^*(\theta) = \frac{x_0}{\phi} H G, \quad H = \gamma(\mu + \gamma v), \quad G = \frac{\mu + \beta v}{\mu + v(\beta - 1)}$$

where $H > 0$ and $G > 0$ by the signs established above. Differentiating,

$$\frac{\partial \tilde{y}^*(\theta)}{\partial \lambda} = \frac{x_0}{\phi} \left[\frac{\partial H}{\partial \gamma} \frac{\partial \gamma}{\partial Q} G + H \frac{\partial G}{\partial \beta} \frac{\partial \beta}{\partial q} p \theta \right]$$

since $\partial Q/\partial \lambda = 1$ and $\partial q/\partial \lambda = p\theta$. In the first term, $\partial H/\partial \gamma = \mu + 2\gamma v = v + \sqrt{(\mu - v)^2 + 4vQ} > 0$ from our previous expression for γ .

Now, $\partial \gamma/\partial Q = 1/(2v\gamma + \mu - v) > 0$, since the denominator is the slope of the convex quadratic $vz^2 + (\mu - v)z - Q$ evaluated at its larger root. The first term is therefore strictly positive.

In the second term, $\partial G/\partial \beta < 0$ and $\partial \beta/\partial q < 0$ from the previous paragraph, so the term is nonnegative, and it is zero only if $p\theta = 0$. Hence $\tilde{y}^*(\theta)$ is strictly increasing in λ . The factor $\phi/(\phi - \alpha S)$ is positive and nondecreasing in λ , since S is increasing in λ , so $y^*(\theta)$ is strictly increasing in λ .

Comparative statics: σ . Volatility enters $y^*(\theta)$ only through $v = \sigma^2/2$, and v enters only through γ and β , since the factor $\phi/(\phi - \alpha S)$ does not involve σ . It therefore suffices to characterize the sign of $\partial \tilde{y}^*/\partial v$. Differentiating the two characteristic equations implicitly

with respect to ν ,

$$\frac{\partial \gamma}{\partial \nu} = \frac{\gamma(1-\gamma)}{2\nu\gamma + \mu - \nu} < 0, \quad \frac{\partial \beta}{\partial \nu} = \frac{\beta(1-\beta)}{2\nu\beta + \mu - \nu} > 0$$

The first is negative because $\gamma > 1$ and its denominator, the slope of the quadratic at its larger root, is positive. The second is positive because $\beta < 0$ makes its numerator negative and its denominator, the slope at the smaller root, is negative as well. Higher volatility pulls γ down and β up.

For this derivative it is convenient to return to the form of $\tilde{y}^*$ that precedes Eq. (A.6) and to separate the roots from the constants. Multiplying numerator and denominator by $q(Q - \mu)(q - \mu)$,

$$\tilde{y}^*(\theta) = \frac{x_0}{\phi} \frac{(Q - \mu)(q - \mu)}{q} \frac{N}{M}, \quad N = \gamma q - \beta Q, \quad M = (\gamma - 1)(q - \mu) + (1 - \beta)(Q - \mu)$$

where only N and M depend on ν . By the quotient rule, $\partial(N/M)/\partial \nu = (N_\nu M - N M_\nu)/M^2$, with $N_\nu = \gamma_\nu q - \beta_\nu Q$ and $M_\nu = \gamma_\nu(q - \mu) - \beta_\nu(Q - \mu)$. Collecting the terms in γ_ν and in β_ν ,

$$N_\nu M - N M_\nu = \gamma_\nu [qM - N(q - \mu)] - \beta_\nu [QM - N(Q - \mu)]$$

Each term simplifies to

$$\begin{aligned} qM - N(q - \mu) &= (Q - q)(q - \beta\mu) \\ QM - N(Q - \mu) &= (Q - q)(Q - \gamma\mu) \end{aligned}$$

The characteristic equations give $q - \beta\mu = \nu\beta(\beta - 1) > 0$ and $Q - \gamma\mu = \nu\gamma(\gamma - 1) > 0$, and $Q - q = \lambda(1 - p\theta)$. Therefore

$$\frac{\partial \tilde{y}^*(\theta)}{\partial \nu} = \frac{x_0}{\phi} \frac{(Q - \mu)(q - \mu)\lambda(1 - p\theta)}{qM^2} \left[(q - \beta\mu) \frac{\partial \gamma}{\partial \nu} - (Q - \gamma\mu) \frac{\partial \beta}{\partial \nu} \right]$$

The term in the brackets is negative, since its first term is negative and its second is positive and enters with a minus sign, and the term multiplying it is positive when $p\theta < 1$. Hence $\tilde{y}^*(\theta)$, and with it $y^*(\theta)$, is strictly decreasing in ν , and so in σ , when $p\theta < 1$. When $p\theta = 1$, $Q = q$, γ and β are the two roots of the same quadratic, and $\tilde{y}^*$ reduces to $x_0 S/\phi = x_0/m_U$, which does not depend on σ . In that case the new investor takes the whole firm at every market round, so the owners value the firm as if it failed at its next financing need, which kills the option value. This completes the proof of Proposition 2. $\square$

A.3. Derivation of the Total Surplus

We show that for any funding decisions $(d_j)_{j \geq 1}$ in $\{0, 1\}$ and any stakes $(s_i)_{i \geq 1}$ in $[0, 1]$, the sum of the discounted payoffs of the founder and of every new investor is

$$U_{0,t} + \sum_{i \geq 1} U_{i,t} = e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} S(t, \tau_\phi) - \sum_{\substack{\text{all rounds } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} x(y_{T_j}) S(t, T_j) \quad (\text{A.7})$$

path by path, which is the integrand of Eq. (8). We write the payoff of each player. The owners together put in $x(y_{T_j})$ at a funded inside round, and the sum of their payoffs does not depend on how they split it. To fix ideas we take the split in an inside round to be in proportion to their holdings, as in Eq. (1).

The founder starts with share 1 of the firm and is diluted by $1 - s_j$ at every funded market round, so at any date u it holds share $D(t, u)$ of the firm. At every funded inside round it contributes $D(t, T_j) x(y_{T_j})$, and if the firm survives to the payout it receives its share of the dividend. Discounting to t ,

$$U_{0,t} = e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} D(t, \tau_\phi) S(t, \tau_\phi) - \sum_{\substack{\text{inside } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} D(t, T_j) x(y_{T_j}) S(t, T_j) \quad (\text{A.8})$$

Investor i is called at market round T_i and receives a payoff only if the firm has survived to T_i and that round is funded, which is the factor $S(t, T_i)$. It then contributes $x(y_{T_i})$ and receives share s_i of the firm. From T_i on it is one of the owners with initial share s_i . It is diluted by later market rounds, so it holds $s_i D(T_i, u)$ at date $u > T_i$, it contributes $s_i D(T_i, T_j) x(y_{T_j})$ at every later funded inside round, and it receives its share of the dividend if the firm survives to the payout. Hence

$$U_{i,t} = S(t, T_i) \left[-e^{-\rho(T_i - t)} x(y_{T_i}) + s_i e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} D(T_i, \tau_\phi) S(T_i, \tau_\phi) - s_i \sum_{\substack{\text{inside } j \\ T_i < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} D(T_i, T_j) x(y_{T_j}) S(T_i, T_j) \right] \quad (\text{A.9})$$

Investors whose round is an inside round never enter, and no investor is called after τ_ϕ , so the sum in Eq. (A.7) runs over the market rounds in $(t, \tau_\phi]$.

At any date u the firm's equity is held by the founder and by every investor whose market

round was funded before u , with shares $D(t, u)$ and $s_i D(T_i, u)$. Therefore we have:

$$D(t, u) + \sum_{\substack{\text{market } i \\ t < T_i \leq u}} s_i D(T_i, u) = 1 \quad (\text{A.10})$$

Adding Eq. (A.8) and Eq. (A.9) over the market rounds and collecting the dividend terms, using $S(t, T_i) S(T_i, \tau_\phi) = S(t, \tau_\phi)$,

$$e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} S(t, \tau_\phi) \left[D(t, \tau_\phi) + \sum_{\substack{\text{market } i \\ t < T_i \leq \tau_\phi}} s_i D(T_i, \tau_\phi) \right] = e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} S(t, \tau_\phi)$$

by Eq. (A.10) at $u = \tau_\phi$.

The remaining terms are

$$\begin{aligned} & - \sum_{\substack{\text{inside } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} D(t, T_j) x(y_{T_j}) S(t, T_j) \\ & - \sum_{\substack{\text{market } i \\ t < T_i \leq \tau_\phi}} \sum_{\substack{\text{inside } j \\ T_i < T_j \leq \tau_\phi}} s_i e^{-\rho(T_j - t)} D(T_i, T_j) x(y_{T_j}) S(t, T_j) \\ & - \sum_{\substack{\text{market } i \\ t < T_i \leq \tau_\phi}} e^{-\rho(T_i - t)} x(y_{T_i}) S(t, T_i) \end{aligned}$$

where in the double sum we used $S(t, T_i) S(T_i, T_j) = S(t, T_j)$. The double sum runs over the pairs of a market round i and a later inside round j . Swapping the order of summation:

$$\begin{aligned} & \sum_{\substack{\text{market } i \\ t < T_i \leq \tau_\phi}} \sum_{\substack{\text{inside } j \\ T_i < T_j \leq \tau_\phi}} s_i e^{-\rho(T_j - t)} D(T_i, T_j) x(y_{T_j}) S(t, T_j) \\ & = \sum_{\substack{\text{inside } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} x(y_{T_j}) S(t, T_j) \sum_{\substack{\text{market } i \\ t < T_i < T_j}} s_i D(T_i, T_j) \end{aligned}$$

Grouping it with the first sum,

$$- \sum_{\substack{\text{inside } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} x(y_{T_j}) S(t, T_j) \left[D(t, T_j) + \sum_{\substack{\text{market } i \\ t < T_i < T_j}} s_i D(T_i, T_j) \right] = - \sum_{\substack{\text{inside } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} x(y_{T_j}) S(t, T_j)$$

by Eq. (A.10) at $u = T_j$. The full cost of every funded inside round appears once.

Collecting the dividend, the inside rounds and the market rounds,

$$U_{0,t} + \sum_{i \geq 1} U_{i,t} = e^{-\rho(\tau_\phi - t)} y_{\tau_\phi} S(t, \tau_\phi) - \sum_{\substack{\text{inside } j \\ t < T_j \leq \tau_\phi}} e^{-\rho(T_j - t)} x(y_{T_j}) S(t, T_j) - \sum_{\substack{\text{market } i \\ t < T_i \leq \tau_\phi}} e^{-\rho(T_i - t)} x(y_{T_i}) S(t, T_i)$$

and the last two sums together run over every round in $(t, \tau_\phi]$, which is Eq. (A.7).

A.4. Proof of Proposition 4

Proof. Below u^* there is no entry, so Eq. (12) becomes

$$\frac{\sigma^2}{2} f''(u) - m f'(u) - (\phi + \lambda) f(u) = 0$$

Its solutions are $e^{\kappa u}$ with κ a root of $\frac{\sigma^2}{2} \kappa^2 - m \kappa - (\phi + \lambda) = 0$, whose roots have opposite signs. The negative root is ruled out, since $e^{\kappa u}$ would then diverge as $u \rightarrow -\infty$ and the mass of live firms below any dividend would be infinite. Hence $f(u) = A e^{\kappa u}$ on $(-\infty, u^*)$ with κ the positive root in Eq. (13) and $A > 0$ a constant pinned down by the solution above u^* . Failures occur at rate λ below u^* , so the total failure flow and the failure flow at dividends the benchmark would fund, those with $u \in [\ln y^*(0), \ln y^*(\theta))$, are

$$\lambda \int_{-\infty}^{\ln y^*(\theta)} A e^{\kappa u} du = \frac{A \lambda}{\kappa} y^*(\theta)^\kappa, \quad \lambda \int_{\ln y^*(0)}^{\ln y^*(\theta)} A e^{\kappa u} du = \frac{A \lambda}{\kappa} [y^*(\theta)^\kappa - y^*(0)^\kappa]$$

Their ratio is Eq. (13). The constant A cancels, which removes η and g , and by Eq. (A.4) the slope α scales $y^*(0)$ and $y^*(\theta)$ by the same factor, so it cancels in the ratio as well. $\square$

A.5. Proof of Proposition 5

Proof. Let $\ell(u)$ be the probability that a firm with log dividend u eventually fails, under a threshold u^* . Over a short interval dt one of three things happens. A financing need arrives, with probability λdt , and the firm fails if $u < u^*$ and otherwise is funded and continues from u . The payout arrives, with probability ϕdt , and the firm never fails. Neither arrives, and the log dividend moves to $u + du$. Hence

$$\ell(u) = \lambda dt \left[\mathbb{1}_{\{u < u^*\}} + \mathbb{1}_{\{u \geq u^*\}} \ell(u) \right] + (1 - (\lambda + \phi) dt) \mathbb{E}[\ell(u + du)]$$

A second-order expansion of $\ell(u + du)$ with $du = m dt + \sigma dZ_t$, using $\mathbb{E}[dZ_t] = 0$ and $\mathbb{E}[dZ_t^2] = dt$ and dropping terms of order dt^2 , gives $\mathbb{E}[\ell(u + du)] = \ell(u) + m \ell'(u) dt + \frac{\sigma^2}{2} \ell''(u) dt$. Substituting,

dividing by dt and taking the limit $dt \rightarrow 0$ gives the following ordinary differential equation for ℓ :

$$0 = \frac{\sigma^2}{2} \ell'' + m \ell' - (\phi + \lambda) \ell + \lambda [\mathbb{1}_{\{u < u^*\}} + \mathbb{1}_{\{u \geq u^*\}} \ell] \quad (\text{A.11})$$

Above u^* , Eq. (A.11) becomes $\frac{\sigma^2}{2} \ell'' + m \ell' - \phi \ell = 0$, whose solutions are e^{zu} with z a root of $\frac{\sigma^2}{2} z^2 + mz - \phi = 0$, one positive and one negative. A firm far above the threshold almost never fails, so $\ell(u) \rightarrow 0$ as $u \rightarrow \infty$ and the positive root is ruled out. Hence

$$\ell(u) = B e^{-\zeta u}, \quad \zeta = \frac{m + \sqrt{m^2 + 2\sigma^2 \phi}}{\sigma^2} > 0$$

where $-\zeta$ is the negative root. Below u^* , Eq. (A.11) becomes $\frac{\sigma^2}{2} \ell'' + m \ell' - (\phi + \lambda) \ell + \lambda = 0$. The constant $\lambda/(\lambda + \phi)$ is a particular solution, and it is the probability that the financing clock rings before the payout clock, which is the probability of failure for a firm so far below the threshold that it never returns. The homogeneous solutions are e^{zu} with z a root of $\frac{\sigma^2}{2} z^2 + mz - (\phi + \lambda) = 0$, and since ℓ is bounded as $u \rightarrow -\infty$, the negative root is ruled out. Hence

$$\ell(u) = \frac{\lambda}{\lambda + \phi} + A e^{r(u-u^*)}, \quad r = \frac{-m + \sqrt{m^2 + 2\sigma^2(\phi + \lambda)}}{\sigma^2} > 0$$

Continuity of ℓ and ℓ' at u^* ¹⁶ gives $\frac{\lambda}{\lambda + \phi} + A = B e^{-\zeta u^*}$ and $rA = -\zeta B e^{-\zeta u^*}$. Eliminating A ,

$$\ell(u) = \frac{\lambda}{\lambda + \phi} \frac{r}{r + \zeta} e^{-\zeta(u-u^*)}, \quad u \geq u^* \quad (\text{A.12})$$

To obtain the failure flow, consider a long horizon T . Over it, $\eta T g(u) du$ firms enter with log dividend in $[u, u + du)$, and each of them eventually fails with probability $\ell(u)$, so the number of firms that fail among those born over the horizon is $\eta T \int g(u) \ell(u) du$. In the stationary equilibrium the failure flow is this number per unit of time, the entry rate times the probability that an entrant eventually fails. Entrants have log dividends above $\ln y^*(\theta)$, where Eq. (A.12) applies, so with the wedge the failure flow is

$$D_\theta = \eta \int_{\ln y^*(\theta)}^{\infty} g(u) \frac{\lambda}{\lambda + \phi} \frac{r}{r + \zeta} e^{-\zeta(u - \ln y^*(\theta))} du$$

Removing the wedge lowers the threshold to $\ln y^*(0)$, which is below the support of g , so Eq. (A.12) still applies to every entrant and the failure flow is the same expression with $\ln y^*(0)$

¹⁶Appendix A.6 shows why ℓ and ℓ' are continuous at the threshold, by approximating the log dividend with a binomial process and taking the limit.

in the exponent, D_0 . The two integrands differ by the constant factor $e^{-\zeta[\ln y^*(\theta) - \ln y^*(0)]}$, so

$$\frac{D_\theta - D_0}{D_\theta} = 1 - \frac{D_0}{D_\theta} = 1 - \left(\frac{y^*(0)}{y^*(\theta)} \right)^\zeta$$

which is Eq. (14). □

A.6. Continuity of ℓ and ℓ' at the Threshold

We approximate the log dividend by a binomial process on a grid and show that the discrete analogues of ℓ and ℓ' are continuous at the threshold without imposing it. Fix a time step Δ , let $h = \sigma\sqrt{\Delta}$, and let the log dividend move from u_i to $u_i + h$ with probability ω and to $u_i - h$ with probability $1 - \omega$, where

$$\omega = \frac{1}{2} + \frac{m}{2\sigma}\sqrt{\Delta}, \quad m = \mu - \frac{\sigma^2}{2}$$

so that the step has mean $m\Delta$ and variance $\sigma^2\Delta$ up to terms of order Δ^2 . Let i^* be the grid index of the threshold, $u_{i^*} = u^*$, and write ℓ_i for the probability that a firm with log dividend u_i eventually fails. Over one step the payout arrives with probability $\phi\Delta$, after which the firm never fails, a financing need arrives with probability $\lambda\Delta$, and otherwise the log dividend moves. Writing $a = (1 - \lambda\Delta)(1 - \phi\Delta)$ and $b = \lambda\Delta(1 - \phi\Delta)$, the law of total probability gives

$$\ell_i = b \mathbb{1}_{\{i < i^*\}} + b \mathbb{1}_{\{i \geq i^*\}} \ell_i + a[\omega \ell_{i+1} + (1 - \omega) \ell_{i-1}] \quad (\text{A.13})$$

since below the threshold a financing need means failure and above it the round is funded and the firm continues from the same state.

For $i < i^*$, Eq. (A.13) is $\ell_i = b + a[\omega \ell_{i+1} + (1 - \omega) \ell_{i-1}]$. A constant $\bar{\ell}$ solves it if $\bar{\ell}(1 - a) = b$, that is,

$$\bar{\ell} = \frac{b}{1 - a} = \frac{\lambda(1 - \phi\Delta)}{\phi + \lambda - \lambda\phi\Delta}$$

which is the probability that the financing clock rings before the payout clock on the grid, the failure probability of a firm so far below the threshold that it never returns, and it converges to $\lambda/(\lambda + \phi)$ as $\Delta \rightarrow 0$. The homogeneous equation has solutions ξ^i with ξ a root of $a\omega\xi^2 - \xi + a(1 - \omega) = 0$. The left side is positive at $\xi = 0$ and equals $a - 1 < 0$ at $\xi = 1$, so the roots satisfy $0 < \xi_-^U < 1 < \xi_+^U$. As $i \rightarrow -\infty$, $(\xi_-^U)^i$ diverges and ℓ_i would not be a probability, so

$$\ell_i^U = \bar{\ell} + B(\xi_+^U)^i$$

for some constant B .

For $i \geq i^*$, Eq. (A.13) is $(1-b)\ell_i = a[\omega\ell_{i+1} + (1-\omega)\ell_{i-1}]$, and its solutions are ξ^i with ξ a root of $a\omega\xi^2 - (1-b)\xi + a(1-\omega) = 0$. The left side is positive at $\xi = 0$ and equals $a - (1-b) = -\phi\Delta < 0$ at $\xi = 1$, so again $0 < \xi_-^F < 1 < \xi_+^F$. As $i \rightarrow \infty$ the failure probability vanishes, which rules out ξ_+^F , so

$$\ell_i^F = A(\xi_-^F)^i$$

for some constant A .

The two branches meet at i^* , and the equations at the two nodes adjacent to the threshold each reach across it. At $i = i^*$ the funded equation is $(1-b)\ell_{i^*}^F = a\omega\ell_{i^*+1}^F + a(1-\omega)\ell_{i^*-1}^U$, and solving for the unfunded value,

$$\ell_{i^*-1}^U = \frac{(1-b)\ell_{i^*}^F - a\omega\ell_{i^*+1}^F}{a(1-\omega)} = A(\xi_-^F)^{i^*-1} \frac{(1-b)\xi_-^F - a\omega(\xi_-^F)^2}{a(1-\omega)} = A(\xi_-^F)^{i^*-1} = \ell_{i^*-1}^F$$

where the third equality uses the characteristic equation of ξ_-^F . At $i = i^* - 1$ the unfunded equation is $\ell_{i^*-1}^U = b + a\omega\ell_{i^*}^F + a(1-\omega)\ell_{i^*-2}^U$, and solving for the funded value,

$$\ell_{i^*}^F = \frac{\ell_{i^*-1}^U - a(1-\omega)\ell_{i^*-2}^U - b}{a\omega} = \frac{\bar{\ell}[1 - a(1-\omega)] - b}{a\omega} + B(\xi_+^U)^{i^*-2} \frac{\xi_+^U - a(1-\omega)}{a\omega} = \bar{\ell} + B(\xi_+^U)^{i^*} = \ell_{i^*}^U$$

The two branches therefore agree at the two adjacent nodes $i^* - 1$ and i^* , so their values and their difference quotients $(\ell_{i^*} - \ell_{i^*-1})/h$ coincide. Neither condition is imposed. Both follow from the recursion at the nodes next to the threshold. As $\Delta \rightarrow 0$ the two branches converge to the two solutions of Eq. (A.11) on either side of u^* , and the two seam conditions become the continuity of ℓ and of ℓ' at u^* used in the proof of Proposition 5. The same argument, applied to the equity value on the grid, gives the continuity of W and W' at y^* used in the proof of Proposition 2.

B Estimation Details

This appendix describes the simulation behind the estimator of Section 5.3. Given a candidate (μ, σ) and the parameters already in hand, the threshold and the equity value are available in closed form from Appendix A.2, so nothing is solved numerically except the inversion of the entry states below. The simulation then produces an event-level panel with the same columns as the data, and the two slopes of Eq. (18) are computed on the simulated panel by the same code that computes them on the data.

B.1. Entry States

The dividend at entry, y_0 , is never observed, but every firm in the estimation sample has an observed stake sold at its first round, s_1 , and the first round is a market round. By Eq. (3), the stake sold at a market round is a strictly decreasing function of the dividend, from one at the threshold toward θ as the dividend grows, since the firm's value rises relative to the constant cost of the round. A firm's entry state is therefore the dividend that reproduces its entry stake, the solution y_0 of

$$s_1 = \theta + (1 - \theta) \frac{x_0}{W(y_0)} \quad (\text{B.1})$$

which we find by solving the equation above numerically. Since y^* and W depend on (μ, σ) , the inversion is repeated at every evaluation of the objective, once for each of the 16,521 firms in the sample. A stake at or below θ cannot be attained at any finite dividend. For the firms whose entry stake falls there, about one percent of the sample, the stake is set just above θ and the entry state is capped at the upper end of the bisection interval, so that the firm is simulated as born far above the threshold.

B.2. Draws

Each firm in the sample is simulated $S = 10$ times, following [Michaelides and Ng \(2000\)](#). A simulated firm is paired with a data firm drawn uniformly with replacement and inherits that firm's entry stake s_1 , hence its entry state, and its observation window $\bar{T}$, the years from its first priced round to the end of the data. The pairing reproduces the right-censoring structure of the data in the simulated panel. All random draws are made once, before the estimation starts, and are kept fixed across candidate parameters, so that the objective is a deterministic function of (μ, σ) and the optimizer changes parameters but not the draws. Table 7 lists the draws behind one simulated firm and what each becomes. The buffer K is chosen so that no simulated firm can exhaust its financing gaps within the longest observation window in the data.

Table 7: The draws behind one simulated firm

Draw	Size	Becomes	Via
$u^{exit} \sim U(0, 1)$	scalar	payout time	$\tau_\phi = -\ln u^{exit}/\phi$
$u_k^{gap} \sim U(0, 1)$	K -vector	gap to financing need k	$\Delta t_k = -\ln u_k^{gap}/\lambda$
$u_k^{type} \sim U(0, 1)$	K -vector	type of round k	market round if $u_k^{type} < p$
$z_k \sim N(0, 1)$	$(K + 1)$ -vector	dividend step to event k	$y_{k+1} = y_k e^{(\mu - \sigma^2/2)\Delta t_k + \sigma\sqrt{\Delta t_k} z_k}$

B.3. Simulation of a Firm and the Panel

A firm is simulated in event time, with one exact step of the geometric Brownian motion between consecutive events rather than a time grid. The firm enters at $t = 0$ with dividend y_0 , and its entry round is recorded with the model's pricing at y_0 , so its stake is s_1 by construction. Its payout time is drawn from u^{exit} . Then, from the current date and dividend, the next financing need arrives after the next gap, and the next event is whichever comes first of three.

- (i) Censoring, when the observation window $\bar{T}$ ends first. The firm's last row is censored and its dividend is not updated.
- (ii) The payout, when τ_ϕ comes first. The dividend is advanced over the elapsed time with the GBM draw, and the firm's last row records the payout with the dividend at that date as its exit value.
- (iii) A financing need, when the gap comes first. The dividend is advanced over the gap. If it is at or above the threshold, the round is funded. The type of the round comes from the uniform draw between 0 and 1. If it is a market round, it is priced by Eq. (3), and otherwise it is an inside round priced at the equity value. The row records the cost of the round, the stake, the post-money valuation and the type of the round, and the loop continues. If the dividend is below the threshold, the firm fails and its last row records the failure.

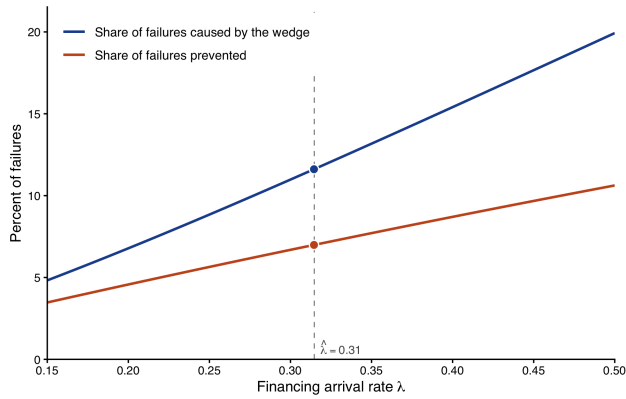
A simulated firm fails only at a financing need that arrives while its dividend is below the threshold, as in the model. Stacking the rows of every simulated firm gives an event-level panel with the same columns as the data panel of Section 4, the entry round, the later rounds with their stakes and valuations, and a terminal row for the payout, the failure or the censoring. The regressions of Eq. (18) are then run on the simulated firms that reach the payout, exactly as they are run on the data firms that exit with a valuation, so the two slopes are subject to the same selection on both sides.

C Sensitivity of the Cost of the Wedge

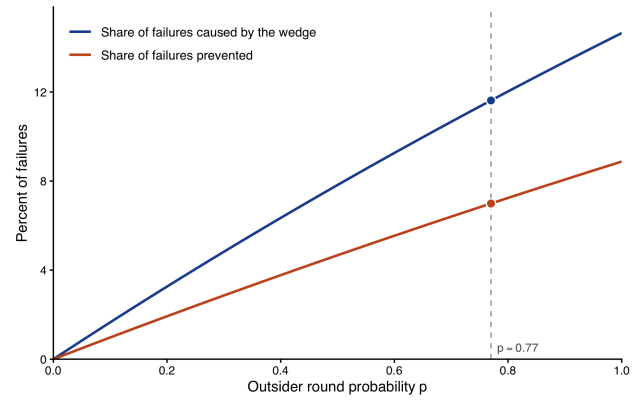
Figure 5 traces the two shares of Propositions 4 and 5 as one parameter at a time varies over a range around its estimate, holding every other parameter at its estimated value, as in Fig. 4. The estimates of μ and σ are not updated along a sweep. Both shares rise with the arrival rate of financing needs and with the probability of a market round, which multiply the rounds at which the surplus leaks to new investors, and fall with the arrival rate of the payout and with the discount rate, which shorten the horizon over which the leakage compounds, the discount rate being the parameter to which the shares are most sensitive.

Figure 5: The cost of the wedge as a function of the other parameters

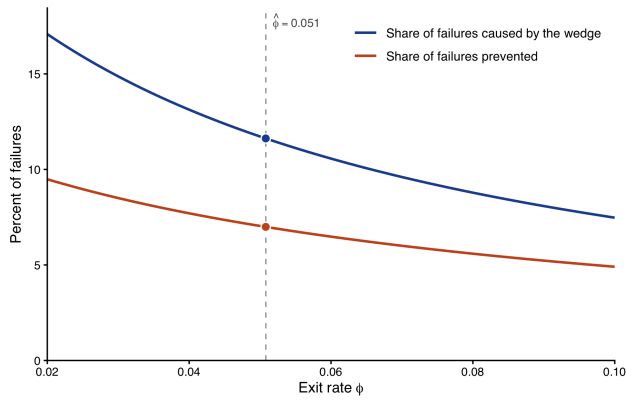
(a) Arrival rate of financing needs λ



(b) Probability of a market round p



(c) Arrival rate of the payout ϕ



(d) Discount rate ρ

